

# Saving by Consuming: The Intertemporal Behavior of Hand-to-Mouth Households<sup>\*</sup>

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\* The experiment developed in this project was registered in the American Economic Association's registry for randomized controlled trials under ID AEARCTR-0012835. The views expressed herein are those of the individual authors and do not necessarily reflect the official positions of the Inter-American Development Bank.

# Background

- ▷ Literature on financial constraints & heterogeneous consumption...
  - Hand-to-Mouth, HtM (Campbell and Mankiw, 1989)
    - ↪ Financial markets exclusion, no forward-looking behavior (Galí et al., 2007)
- ▷ However, many goods have a built-in intertemporal dimension (Attanasio, 1999)
  - Durables enable intertemporal choices while consuming, even if not saving
- ▷ HtM are a sizable portion of the population,  $\approx 40\%$  in the U.S. (Aguiar et al., 2025)
  - “Getting the micro foundations right” essential for aggregate dynamics
- ▶ This paper:
  - Revisit the role of the HtM agent by allowing durable consumption
    - ↪ Exhibit forward-looking behavior
    - ↪ Expectations matter for individual & aggregate outcomes

# This paper: Implications of HtM forward-looking behavior

## 1. Partial equilibrium model: HtM & durables

- HtM smooths consumption, reallocates spending within-period/ across goods
  - ↪ Durable as a saving vehicle; expectations shape consumption choices
- Expectations shifts: different predictions for HtM/non-HtM/myopic HtM

## 2. RCT: designed a novel module on Cleveland FED's SCE

- Causally estimate consumption responses to changes in inflation expectations
  - ↪ Exogenous variation by randomized treatment exposure
- Results validate predictions of PE model

## 3. Incorporate durables into a two-agent New Keynesian (TANK) model

- Spending becomes more volatile, consumption becomes more stable
- Durables dampen the role of household heterogeneity

# Contributions

1. Consumption responses to expectations Bachmann et al. (2015); Armantier et al. (2016); Cavallo et al. (2017); Roth and Wohlfart (2020); D'Acunto et al. (2021); Coibion et al. (2022, 2023); Roth et al. (2023); Andrade et al. (2023); Coibion et al. (2024); Candia (2024); Hajdini et al. (2024); Jiang et al. (2024); Aktug and Atesagaoglu (2024)
  - Causal evidence linking inflation expectations & consumption decisions on durable categories for HtM & non-HtM consumers
2. The HtM agent & TANK models Campbell and Mankiw (1989); Zeldes (1989); Alvarez et al. (2001); Galí et al. (2007); Bilbiie (2008, 2020); Bilbiie et al. (2012); Kaplan and Violante (2014); Kaplan et al. (2014); Debortoli and Galí (2024); Aguiar et al. (2025), & durables Attanasio (1999); Caballero (1993); Barsky et al. (2007, 2016); McKay et al. (2016); McKay and Wieland (2021, 2022); Beraja and Wolf (2021); Bachmann et al. (2024)
  - Show that durability introduces a forward-looking behavior for HtM agents
  - Extend the TANK framework with durables, redefining HtM agents' role in GE models

# Two-agent, two-good framework

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# Two households & two goods | Key ingredients

- ▷ Two infinitely lived households:  $i = \{H, S\}$ 
  - HtM,  $i = H$ : excluded from financial markets (Galí et al., 2007; Bilbiie, 2008)
  - non-HtM,  $i = S$ : access to financial markets
- ▷ Two goods:  $j = \{d, c\}$ 
  - $c_{i,t}$ : non-durable consumption in  $t$
  - $d_{i,t}$ : stock of durables at the end of  $t$
- ▷ Law of motion:  $d_{i,t} = z_{i,t} + (1 - \delta) d_{i,t-1}$ 
  - $\delta$ : depreciation rate
  - $z_{i,t}$ : new durables acquisition in  $t$ , reversibility
- ▷ One-period, risk-free bond:  $B_t$ , only for non-HtM
  - $R_{t+1}$ : gross nominal return, no-arbitrage condition

# Households' optimization problem

▷ HtM solves

$$\begin{aligned} & \max_{\{z_{i,k}, c_{i,k}\}_{k \geq t}} \sum_{k \geq t} \beta^{k-t} \mathbb{E}_t [u(c_{i,k}, d_{i,k})] \\ \text{s.t.} \quad & P_{c,t} c_{i,t} + P_{d,t} z_{i,t} = P_{w,t} \\ & d_{i,t} = z_{i,t} + (1 - \delta) d_{i,t-1} \end{aligned}$$

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$$\max_{\{z_{i,k}, c_{i,k}\}_{k \geq t}} \sum_{k \geq t} \beta^{k-t} \mathbb{E}_t [u(c_{i,k}, d_{i,k})]$$

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▷ non-HtM solves

$$\max_{\{z_{i,k}, c_{i,k}, \quad \}_{k \geq t}} \sum_{k \geq t} \beta^{k-t} \mathbb{E}_t [u(c_{i,k}, d_{i,k})]$$

$$\begin{aligned} \text{s.t.} \quad & P_{c,t} c_{i,t} + P_{d,t} z_{i,t} = P_{w,t} \\ & d_{i,t} = z_{i,t} + (1 - \delta) d_{i,t-1} \end{aligned}$$



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▷ non-HtM solves

$$\begin{aligned} & \max_{\{z_{i,k}, c_{i,k}, B_k\}_{k \geq t}} \sum_{k \geq t} \beta^{k-t} \mathbb{E}_t [u(c_{i,k}, d_{i,k})] \\ \text{s.t.} \quad & P_{c,t} c_{i,t} + P_{d,t} z_{i,t} = P_{w,t} + (R_t B_{t-1} - B_t) \\ & d_{i,t} = z_{i,t} + (1 - \delta) d_{i,t-1} \end{aligned}$$

# Households' optimization problem

▷ **HtM** solves

$$\begin{aligned} & \max_{\{z_{i,k}, c_{i,k}\}_{k \geq t}} \sum_{k \geq t} \beta^{k-t} \mathbb{E}_t [u(c_{i,k}, d_{i,k})] \\ \text{s.t.} \quad & P_{c,t} c_{i,t} + P_{d,t} z_{i,t} = P_{w,t} \\ & d_{i,t} = z_{i,t} + (1 - \delta) d_{i,t-1} \end{aligned}$$

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$$\begin{aligned} & \max_{\{z_{i,k}, c_{i,k}, B_k\}_{k \geq t}} \sum_{k \geq t} \beta^{k-t} \mathbb{E}_t [u(c_{i,k}, d_{i,k})] \\ \text{s.t.} \quad & P_{c,t} c_{i,t} + P_{d,t} z_{i,t} = P_{w,t} + (R_t B_{t-1} - B_t) \\ & d_{i,t} = z_{i,t} + (1 - \delta) d_{i,t-1} \end{aligned}$$

▷ **Nomeraire**: non-durable,  $P_{c,t}$  =  $\left\{ \begin{array}{l} q_t = P_{d,t}/P_{c,t} : \text{relative price of durables} \\ w_t = P_{w,t}/P_{c,t} : \text{real wage} \\ r_t = R_t/\pi_{c,t} : \text{real interest rate} \end{array} \right.$   
 ( $\pi_{c,t} = P_{c,t}/P_{c,t-1}$ : gross inflation)

# Dual nature: consumption good & storable asset

▷ Example: two future states,  $t + 1$  onward

- High-income:  $P_{w,k} = Y_h$  for  $k \geq t + 1$ , with probability  $p$
- Low-income:  $P_{w,k} = Y_l < Y_h$ , with probability  $(1 - p)$

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- Low-income:  $P_{w,k} = Y_l < Y_h$ , with probability  $(1 - p)$

▷ Trade-off: two benefits offset the cost of reducing  $c_{i,t}$

- Marginal increase in  $d_{i,t}$ , immediate utility
- Undepreciated marginal  $d_{i,t}$ , discounted expected future utility

$$mu_{H,1}^c = mu_{H,1}^d + \beta (1 - \delta) [p \cdot mu_{H,2}^d(Y_h) + (1 - p) \cdot mu_{H,2}^d(Y_l)]$$

▷ Durability intertemporal benefits: bias the decision towards  $d_{i,t}$

- Beliefs ( $p$ ) shape consumption choices
- Stronger incentives for accumulation in  $Y_l$  ( $p = 0$ ) scenario
- Saving vehicle  $d_{i,t}$ : smooth consumption despite lack of bonds

# Inflation expectations | Optimal consumption responses

- ▷ One-time/isolated/exogenous change in expected inflation
- ▷ Households learn about inflation expectations changes one period ahead,  $\mathbb{E}_t [\pi_{j,t+1}]$
- ▷ What are the optimal consumption responses?

$$\frac{\partial z_{i,t}}{\partial \mathbb{E}_t [\pi_{j,t+1}]} \quad \& \quad \frac{\partial c_{i,t}}{\partial \mathbb{E}_t [\pi_{j,t+1}]} \quad \text{with } i = \{H, S\}, \quad j = \{d, c, w\}$$

- Policy function: household-specific
- Responses hinge on changes in expected relative prices:  $\mathbb{E}_t [q_{t+1}]$ ,  $\mathbb{E}_t [w_{t+1}]$  &  $\mathbb{E}_t [r_{t+1}]$
- HtM: no sensitivity to changes in expected real interest rate

# No durables, canonical TANK | Myopic HtM

## Proposition 1

If durables fully depreciate ( $\delta = 1$ ), HtM is unresponsive to changes in inflation beliefs

$$\left. \frac{\partial z_{H,t}}{\partial \mathbb{E}_t [\pi_{j,t+1}]} \right|_{\delta=1} = \left. \frac{\partial c_{H,t}}{\partial \mathbb{E}_t [\pi_{j,t+1}]} \right|_{\delta=1} = 0$$

By contrast, the non-HtM remains responsive even when  $\delta = 1$ , adjusting its consumption of every good in the same direction.

- Without durables, HtM expectations play no role
- non-HtM expectations matter: intertemporal substitution via financial markets

## Durables, $\delta < 1$ | Expected income shock

### Proposition 2

Given a one-time, isolated and exogenous change in the expected inflation of the nominal wage  $\mathbb{E}_t [\pi_{w,t+1}]$ , the HtM responses satisfy

$$\frac{\partial z_{H,t}}{\partial \mathbb{E}_t [\pi_{w,t+1}]} < 0 \quad \& \quad \frac{\partial c_{H,t}}{\partial \mathbb{E}_t [\pi_{w,t+1}]} > 0$$

By contrast, given the non-arbitrage condition, the non-HtM responses satisfy

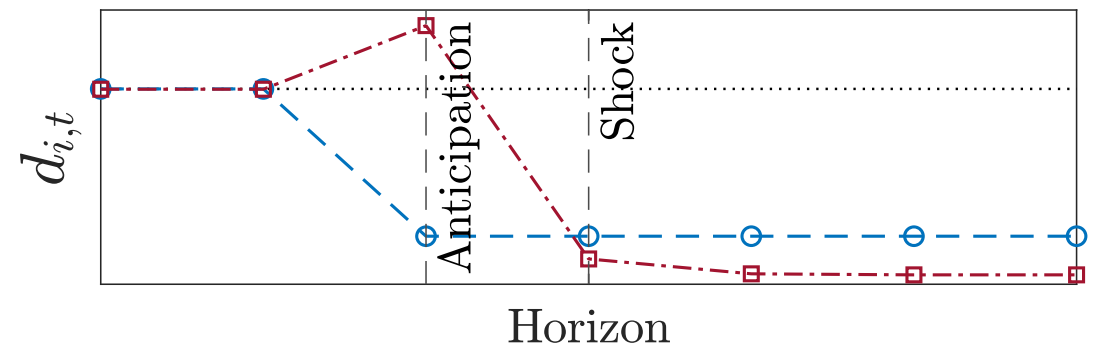
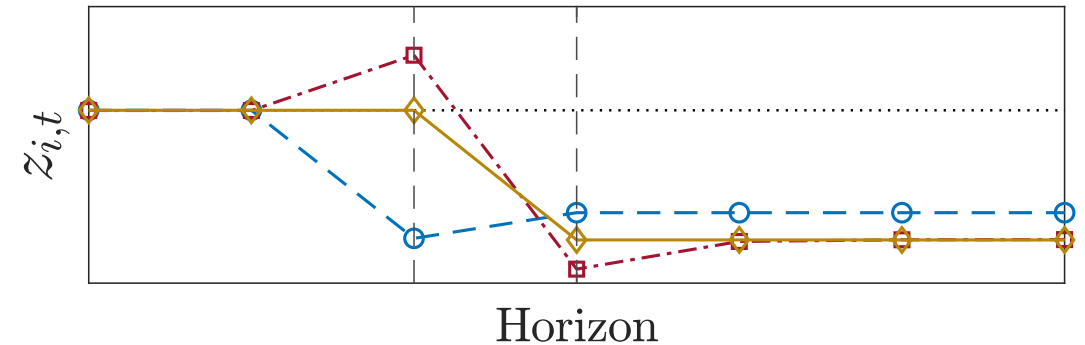
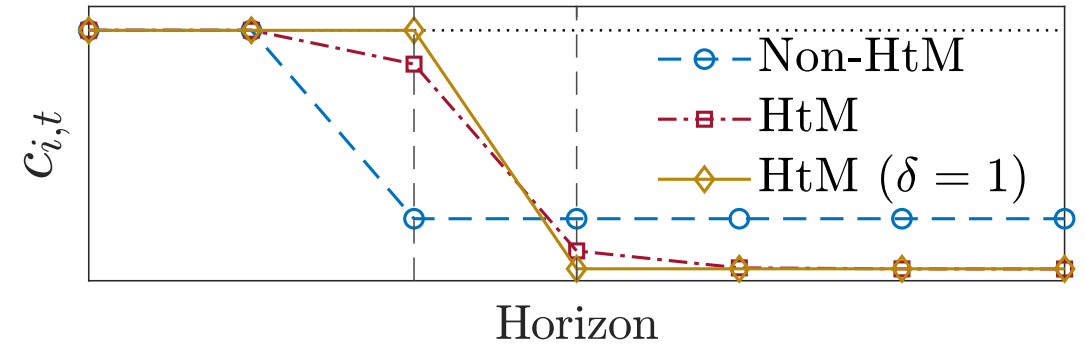
$$\frac{\partial z_{S,t}}{\partial \mathbb{E}_t [\pi_{w,t+1}]} > 0 \quad \& \quad \frac{\partial c_{S,t}}{\partial \mathbb{E}_t [\pi_{w,t+1}]} > 0 \quad \& \quad \frac{\partial b_{S,t}}{\partial \mathbb{E}_t [\pi_{w,t+1}]} < 0$$

- To smooth consumption, HtM use durables as a borrowing instrument

# Expected income shock

▷ Lower expected wage inflation

- **non-HtM**: Permanent Income Hypothesis
- **Forward-looking HtM**: wealth effect accelerates **durable accumulation** (saving instrument)
- **Myopic HtM** ( $\delta = 1$ ): **unresponsive**





# Inflation expectations shocks | Predictions

▷ **Prediction 1:** ↑ aggregate inflation expectations = ↓ expected real wages & ↓ real rate

▷

|           | Higher aggregate inflation expectations |             |  |
|-----------|-----------------------------------------|-------------|--|
|           | Durable                                 | Non-durable |  |
| Household |                                         |             |  |
| HtM       |                                         |             |  |
| non-HtM   |                                         |             |  |

# Inflation expectations shocks | Predictions

▷ **Prediction 1:** ↑ aggregate inflation expectations = ↓ expected real wages & ↓ real rate

▷

| Household | Higher aggregate inflation expectations |             |  |
|-----------|-----------------------------------------|-------------|--|
|           | Durable                                 | Non-durable |  |
| HtM       | Up                                      | Down        |  |
| non-HtM   |                                         |             |  |

# Inflation expectations shocks | Predictions

▷ **Prediction 1:** ↑ aggregate inflation expectations = ↓ expected real wages & ↓ real rate

▷

| Household | Higher aggregate inflation expectations |             |  |
|-----------|-----------------------------------------|-------------|--|
|           | Durable                                 | Non-durable |  |
| HtM       | Up                                      | Down        |  |
| non-HtM   | Up* / Down                              | Up* / Down  |  |

# Inflation expectations shocks | Predictions

- ▷ **Prediction 1:** ↑ aggregate inflation expectations = ↓ expected real wages & ↓ real rate
- ▷ **Prediction 2:** ↑ durable inflation expectations = ↑ expected relative price of durables

|           | Higher aggregate inflation expectations |             | Higher durable inflation expectations |             |
|-----------|-----------------------------------------|-------------|---------------------------------------|-------------|
| Household | Durable                                 | Non-durable | Durable                               | Non-durable |
| HtM       | Up                                      | Down        | Up                                    | Down        |
| non-HtM   | Up* / Down                              | Up* / Down  | Up                                    | Up* / Down  |

# Data

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# Data source | Cleveland Fed's Survey of Consumer Expectations

## ▷ Novel module

- Designed by us for this specific experiment
- Integrated into the Cleveland Fed's SCE (Knotek II et al., 2020; Dietrich et al., 2022)

## ▷ Key features

- Repeated cross-section with daily frequency
- Online U.S. representative sample restricted to +18 years old

## ▷ Sample: $\approx$ 5,500 valid observations post-filters (Haaland et al., 2023)

## ▷ Collected between November 21, 2023, & February 13, 2024

## ▶ **RCT**: information treatments to generate exogenous variation in expectations

# Experimental setup | Timeline

**I. Collect baseline information:** (1) background / **HtM status**, (2) past consumption expenditure, & (3) short-run inflation expectations / **prior**



**II. Apply treatments:** 1 passive control group & 1 placebo & 2 **information treatments**



**III. Collect post-treatment information:** medium-run inflation expectations / **posterior**



**IV. Collect post-treatment information:** intended **consumption expenditure**

# HtM candidates | Who are the HtM?

- ▷ Identifying HtM respondents is challenging
- ▷ Our approach: **key indicators of HtM status**
  - Self-perception using “*living paycheck-to-paycheck*”
  - Self-perception of **financial constraints**
  - Effective **financial market participation**
  - **Liquid assets** relative to consumption
- ▷ Dimension reduction: PCA to synthesize multivariate information
  - 1st component eigenvalue  $\gg$  Kaiser threshold; captures  $\approx 60\%$  of variance
  - Captures dominant and significant share of common variation
- ▶ **HtM: bottom 40%** (baseline) of 1st component distribution, robustness 30%-50% (Campbell and Mankiw, 1989; Kaplan and Violante, 2014; Kaplan et al., 2014; Aguiar et al., 2025)

▶ HtM questions

▶ Scatterplot & scree plot

▶ Demographics by HtM status



# Inflation expectations | Prior beliefs

- ▷ Quantitative point forecast: accurate on first-moment predictions (Clements, 2010, 2014)
- ▷ Established two-step framework to elicit point forecast (Armantier et al., 2017)

## Prior : short-run inflation expectations

Q1.1 Over the next 12 months, do you think that there will be inflation or deflation?

- ☐ Inflation
- ☐ Deflation (the opposite of inflation)

As a follow-up question...

↪ What do you expect the rate of <inflation> <deflation> to be over the next 12 months?

Please give your best guess.

- I expect the rate of <inflation> <deflation> to be \_\_\_\_\_ percent over the next 12 months.

# Information treatments | Random assignment

## **Treatment 1** : *expected aggregate inflation*

T1 According to the Survey of Professional Forecasters conducted by the [Federal Reserve Bank of Philadelphia](#), the annual aggregate inflation rate at the end of the year 2024 will be [2.5%](#). / [1,410 observations](#)

## **Treatment 2** : *past semi-durables inflation*

T2 According to the [Bureau of Labor Statistics](#), between 2012 and 2019, average apparel prices stayed relatively constant. On average, these prices decreased by 0.24% annually. In the last 12 months, average apparel prices have increased by [3.1%](#). / [1,354 observations](#)

## **Placebo** : *past population growth*

P According to the [Census](#), the U.S. population grew [2.9%](#) between 2018 and 2022. / [1,370 observations](#)

- **Passive control group** / [1,395 observations](#)

# Inflation expectations | Posterior beliefs

▷ Horizon difference to avoid question repetition bias (Haaland et al., 2023)

## Posterior : *medium-run inflation expectations*

Q1.2 Over the next 5 years, do you think there will be inflation or deflation on average?

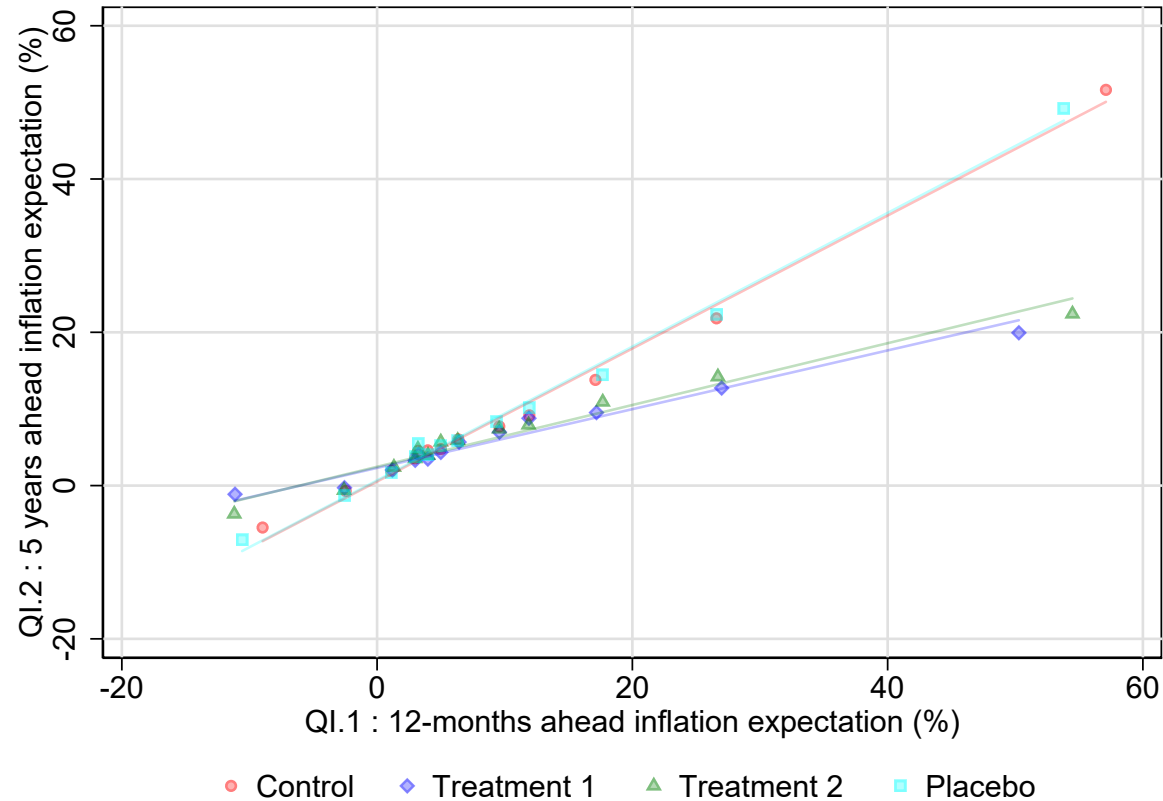
- ☐ Inflation.
- ☐ Deflation (the opposite of inflation).

As a follow-up question...

- ↪ What do you expect the average annual rate of  $\langle \text{inflation} \rangle$   $\langle \text{deflation} \rangle$  to be over the next 5 years? Please give your best guess.
- I expect the average annual rate of  $\langle \text{inflation} \rangle$   $\langle \text{deflation} \rangle$  to be \_\_\_\_\_ percent per year over the next 5 years.

# Treatments | Validation

Prior v/s posterior beliefs by treatment status



- Treatments moderate expectations, similarly across treated groups
- Placebo indistinguishable from control group

# Treatments | Validation

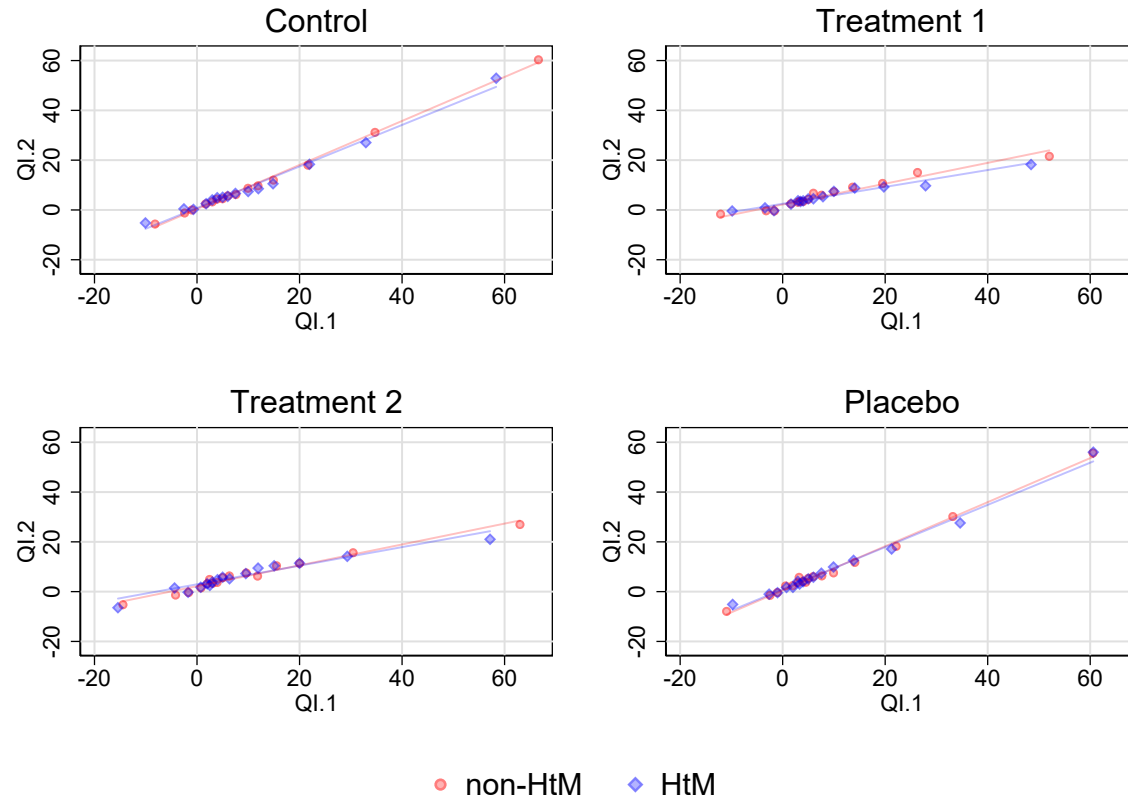
▷ To evaluate treatment effects on medium-run expectations

$$\textit{post}_j = \alpha + \mathbf{X}_j \beta' + \eta \cdot \textit{prior}_j + \sum_{\tau=\{1,2,P\}} [\gamma_{\tau,1} \cdot \textit{treat}_{j,\tau} + \gamma_{\tau,2} (\textit{treat}_{j,\tau} \times \textit{prior}_j)] + v_j$$

- $\textit{prior}_j$ : 12-month inflation expectations of respondent  $j$
- $\textit{post}_j$ : 5-year average annual inflation expectations
- $\textit{treat}_{j,\tau} = \{0, 1\}$ : 1 if  $j$  received the treatment  $\tau = \{1, 2\}$  or placebo P, 0 otherwise
- $\mathbf{X}_j$ : baseline respondent-level covariates, measured before treatment
- **Key coefficients:**  $\gamma_{\tau,2}$  captures learning from treatment  $\tau$

# Treatments across HtM status | Validation

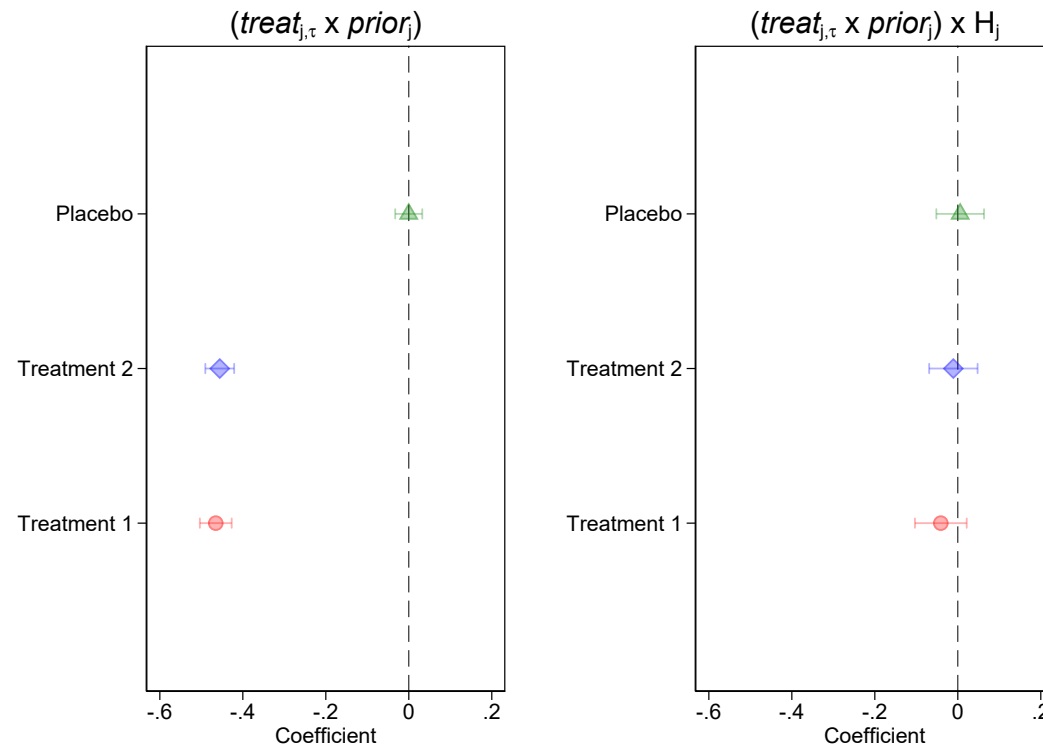
Prior v/s posterior beliefs by treatment and HtM status



- HtM and non-HtM respondents show **similar slopes** across treatment groups
- Suggest **no significant differences in beliefs updating** by HtM status

# Treatments | Two key results

Treatment effects on medium-run inflation expectations



- Placebo: no effect on belief updating / T1 & T2: significant revision
- HtM v/s non-HtM: no differences in treatment assimilation

# Past & intended consumption | Survey questions

- ▷ Survey spending plans track actual behavior (Schnorpfeil et al., 2023; Kotsogiannis and Sakellaris, 2025)

## Intended consumption : *total spending*

QC.1 Compared with your spending **last month**, how do you expect your **total spending** to change in the next ... 1. One month? / 2. One year?

- ☐ Go up by \_\_\_\_\_ %.
- ☐ No change.
- ☐ Go down by \_\_\_\_\_ %.

QC.2 **Services**: *medical and dental care, haircuts, and restaurant meals...*

QC.3 **Non-durables**: *food and beverage products from grocery stores, and gasoline and other energy goods...*

QC.4 **Semi-durables**: *clothing and footwear, jewelry, watches, silverware, toys, tools and garden equipment, and household textiles and utensils...*

QC.5 **Durables**: *household appliances, radio and television sets, sporting and wheeled goods...*



# Empirical strategy & results

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# Identification strategy, no heterogeneity | 2SLS

▷ First-stage: expectations updating

$$post_j = \alpha + \mathbf{X}_j \boldsymbol{\beta}' + \eta \times prior_j + \sum_{\tau=\{1,2\}} [\gamma_{\tau,1} \times treat_{j,\tau} + \gamma_{\tau,2} (treat_{j,\tau} \times prior_j)] + v_j$$

- $prior_j$ : 12-month inflation expectations of respondent  $j$
- $post_j$ : 5-year average annual inflation expectations
- $treat_{j,\tau} = \{0, 1\}$ : 1 if  $j$  received the treatment  $\tau = \{1, 2\}$ , and 0 otherwise

# Identification strategy, no heterogeneity | 2SLS

▷ First-stage: expectations updating

$$\text{post}_j = \alpha + \mathbf{X}_j \beta' + \eta \times \text{prior}_j + \sum_{\tau=\{1,2\}} [\gamma_{\tau,1} \times \text{treat}_{j,\tau} + \gamma_{\tau,2} (\text{treat}_{j,\tau} \times \text{prior}_j)] + v_j$$

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- $\text{treat}_{j,\tau} = \{0, 1\}$ : 1 if  $j$  received the treatment  $\tau = \{1, 2\}$ , and 0 otherwise

▷ Second-stage: consumption responses

$$\% \Delta c_{j,k,h}^e = \alpha_{k,h} + \mathbf{X}_j \beta'_{k,h} + \eta_{k,h} \times \text{prior}_j + \theta_{k,h} \times \widehat{\text{post}}_j + \varepsilon_{j,k,h}$$

- $\% \Delta c_{j,k,h}^e$ : respondent  $j$ 's intended expenditure growth on category  $k$  at horizon  $h$
- **Key coefficients:**  $\theta_{k,h}$  captures the expectations' causal effect on  $\% \Delta c_{j,k,h}^e$

# Heterogeneous effects by HtM status | 2SLS

▷ Adapt the first-stage

- HtM identifier  $H_j = \begin{cases} 1 & \text{if respondent } j \text{ is HtM} \\ 0 & \text{otherwise} \end{cases}$
- Two household-type-specific variables to instrumentalize

$$[ \text{post}_j \times (1 - H_j) , \text{post}_j \times H_j ]$$

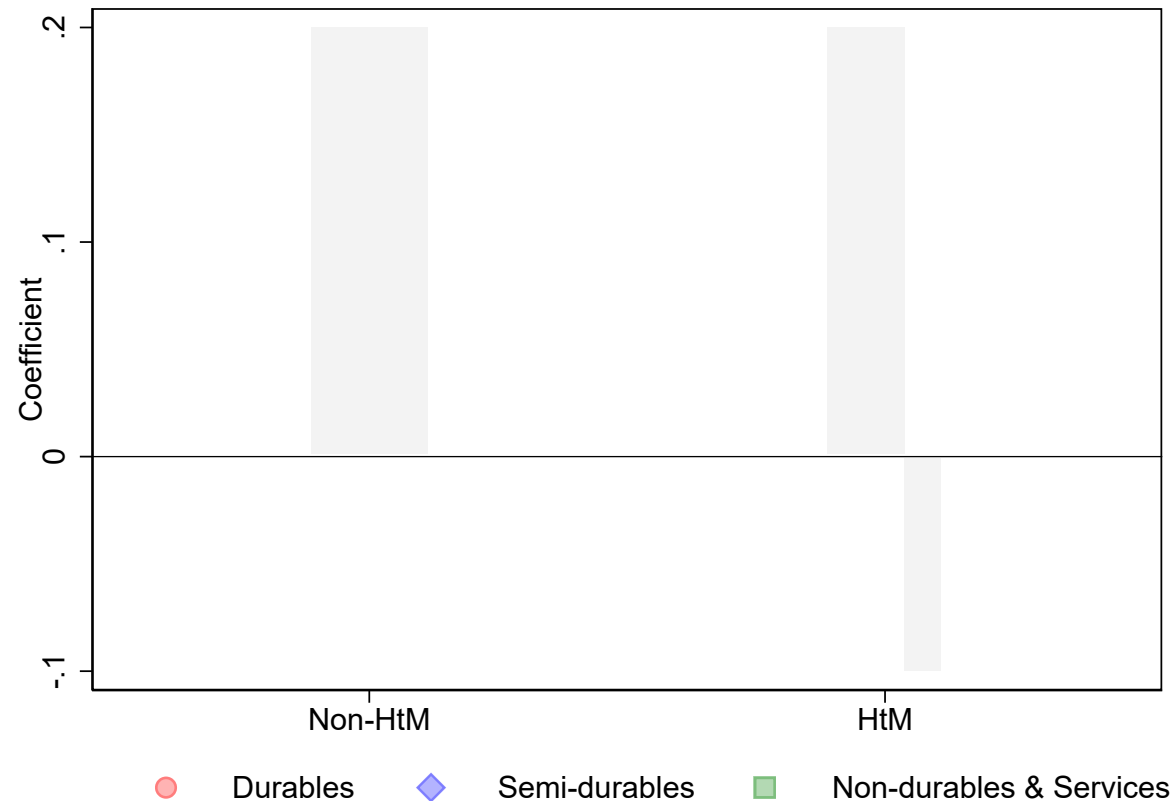
▷ Second-stage:

$$\% \Delta c_{j,k,h}^e = \alpha_{k,h} + \mathbf{X}_j \beta'_{k,h} + \eta_{k,h} \times \text{prior}_j + \theta_{k,h}^S \times \text{post}_j \times (1 - H_j) + \theta_{k,h}^H \times \text{post}_j \times H_j + \varepsilon_{j,k,h}$$

- **Key coefficients:**  $\theta_{k,h}^i$  captures the expectations' causal effect for  $i = \{H, S\}$

# PE | Prediction 1: Higher aggregate inflation expectations

Aggregate inflation expectations changes on consumption by HtM status



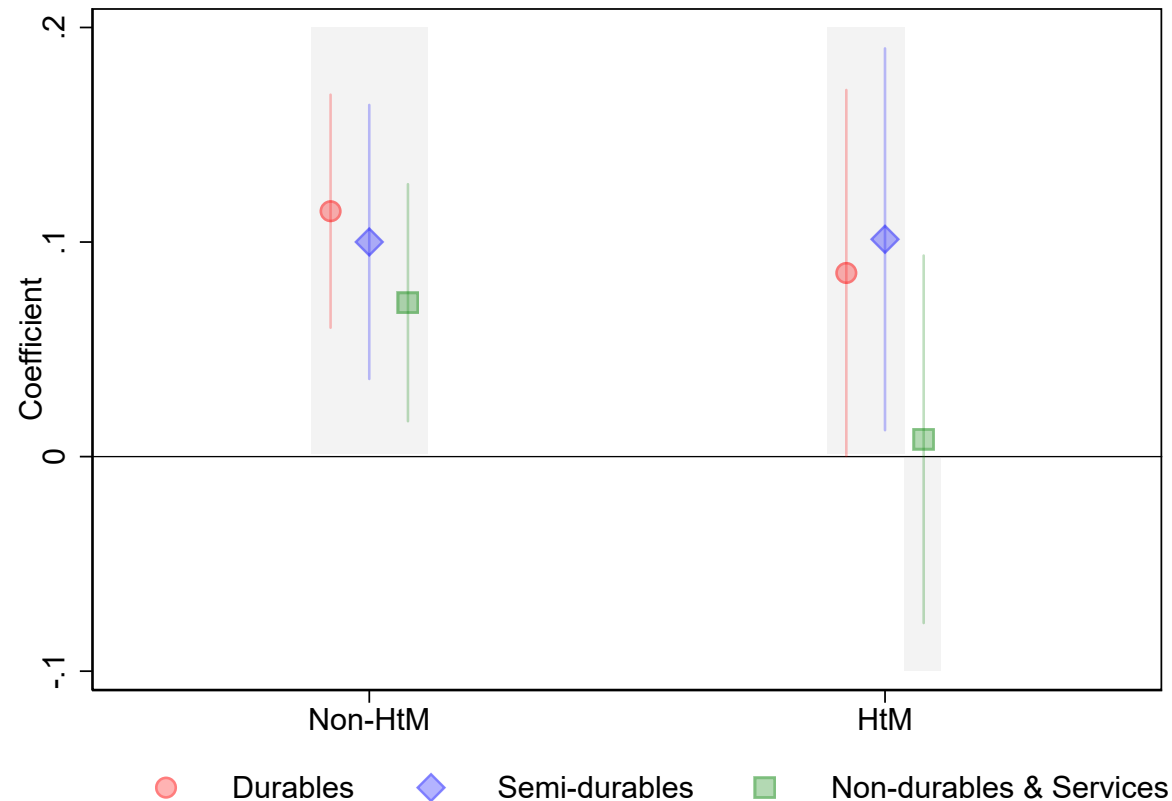
- Non-HtM: increases spending in **all consumption categories**
- HtM: increases **durable investment**, sacrificing non-durable expenditure

► Full results

► Sensitivity

# Evidence | Higher aggregate inflation expectations, pooled

Aggregate inflation expectations changes on consumption by HtM status



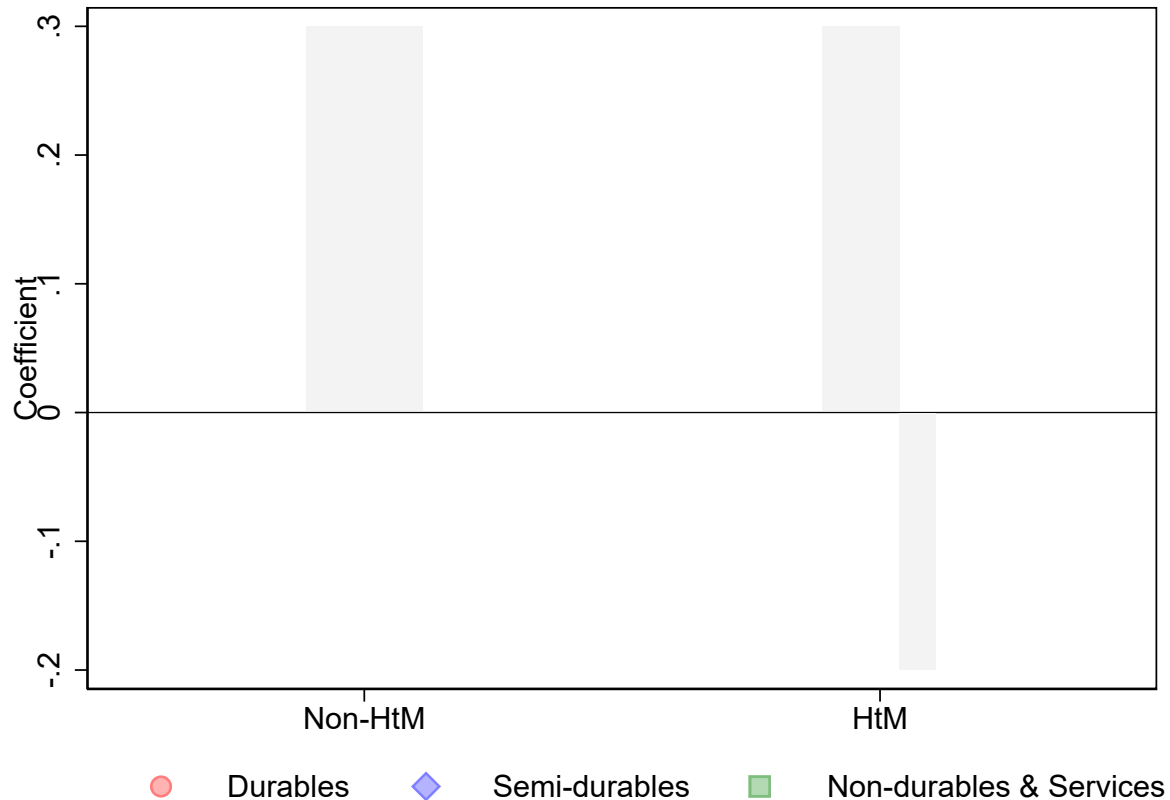
- Non-HtM: increases spending in **all consumption categories**
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► Full results

► Sensitivity

# PE | Prediction 2: Higher durable inflation expectations

Durable inflation expectations changes on consumption by HtM status

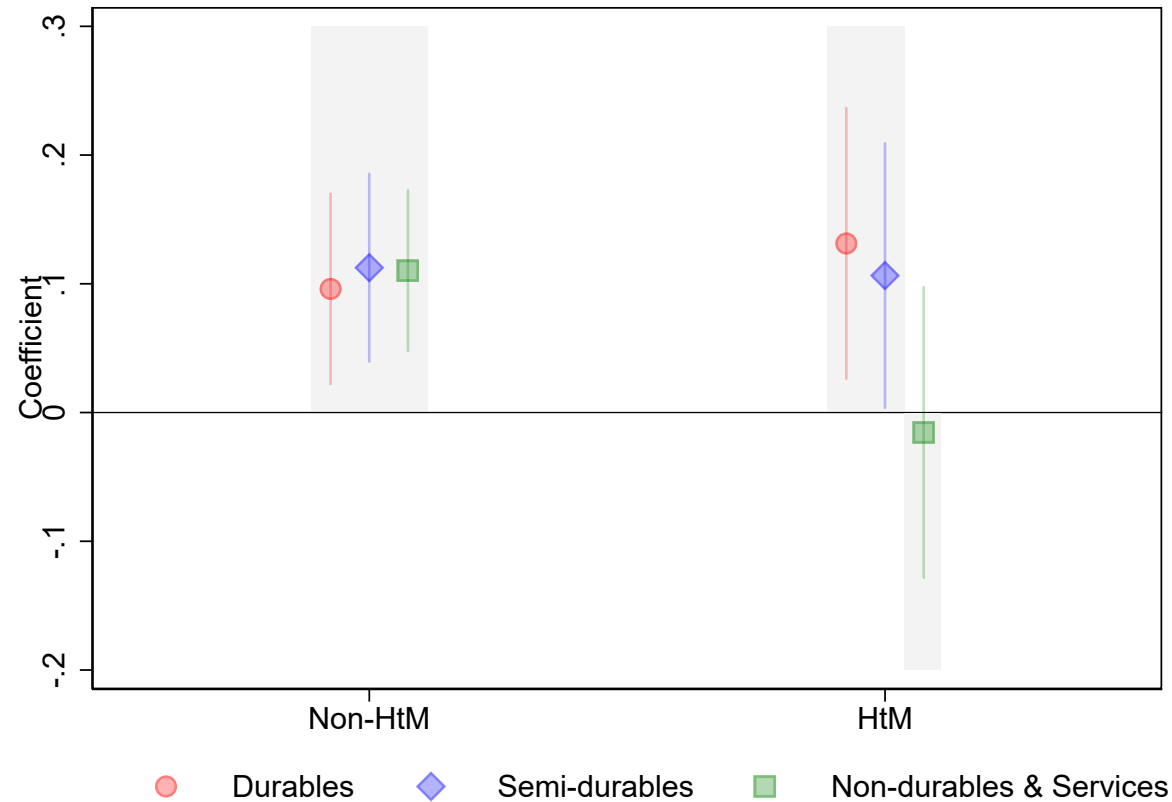


- Non-HtM: increases spending in **all consumption categories**
- HtM: increases **durable investment**, sacrificing non-durable expenditure

► [Full results](#)

# Evidence | Higher durable inflation expectations, T2

Durable inflation expectations changes on consumption by HtM status



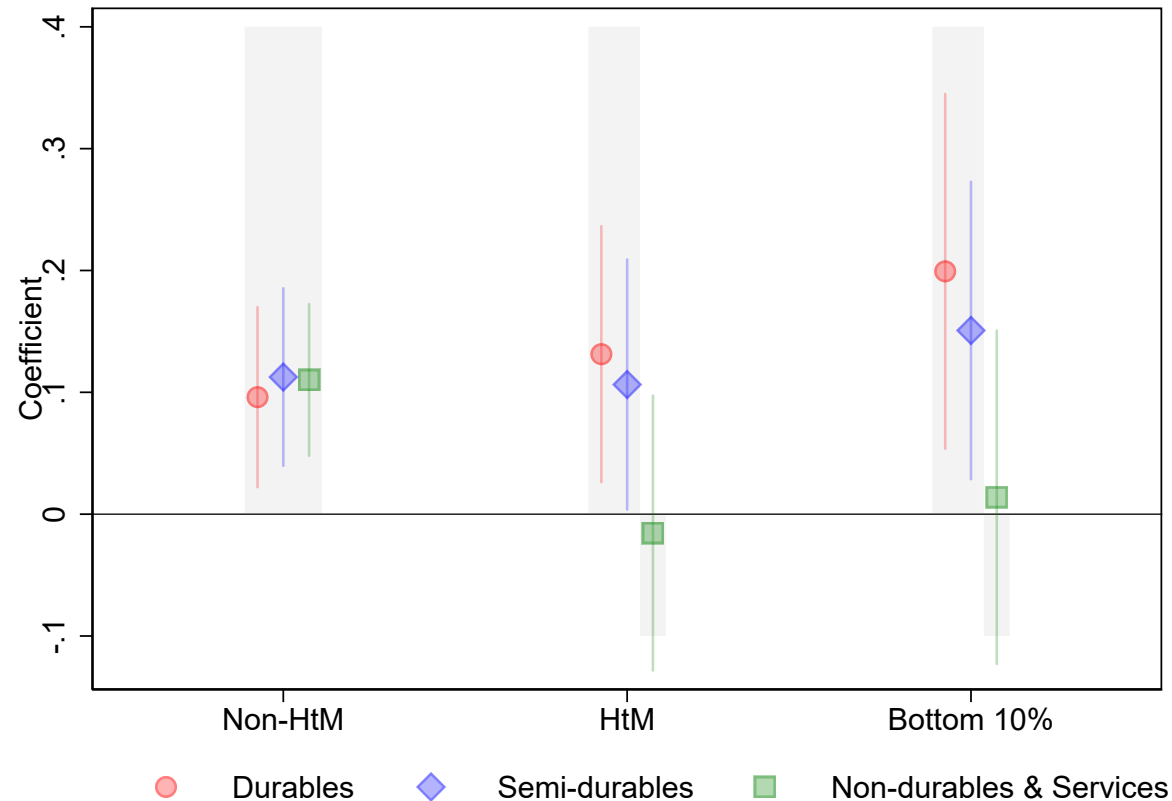
- Non-HtM: increases spending in **all consumption categories**
- HtM: increases **durable investment**, sacrificing non-durable expenditure

► [Full results](#)



# Evidence | Higher durable inflation expectations, T2

Durable inflation expectations changes on consumption of most restricted HtM



- **Most restricted HtM:** financial constraints amplify adjustments; relevance of durables as a smoothing mechanism for most vulnerable households

# General equilibrium analysis

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# Model sketch

- ▶ **Environment:** textbook TANK + durable industry
- ▷ Two **representative households** consuming durables & non-durables
- ▷ Rest of the economy
  - Labor-only production of intermediate goods, under nominal rigidities
  - Abstract from zero-net-supply in the market of bonds
  - Nominal interest rate set by a monetary authority

# Households | Overview

- ▷ Unit mass of infinitely lived households with identical preferences
  - HtM ( $i = H$ ): share  $\Lambda^H \in (0, 1)$
  - non-HtM ( $i = S$ ): share  $\Lambda^S = (1 - \Lambda^H)$
- ▷ Identical situation at the household-type-level: two representative agents
- ▷ Different from the PE model
  - Endogenous labor supply
  - non-HtM acquires equity shares (Bilbiie, 2008)
    - ↪  $\Omega_{j,t-1}$ : shares in sector  $j = \{c, d\}$  at the beginning of  $t$
    - ↪  $v_{j,t}$ : average market value of  $\Omega_{j,t}$
    - ↪  $P_{j,t}\Pi_{j,t}$ : nominal profits per share

# Households | Optimization problem

▷ Representative household  $i = \{H, S\}$  solves

$$\max_{\{z_{i,k}, c_{i,k}, l_{i,k}, \{B_k, \{\Omega_{j,k}\}_{j=\{d,c\}}\} \cdot \mathbb{1}_{i=S}\}_{k \geq t}} \sum_{k \geq t} \beta^{k-t} \mathbb{E}_t [u(c_{i,k}, d_{i,k}, l_{i,k})]$$

$$\begin{aligned} \text{s.t.} \quad P_{c,t} c_{i,t} + P_{d,t} z_{i,t} &= P_{w,t} l_{i,t} + T_{i,t} + (R_t B_{t-1} - B_t) \cdot \mathbb{1}_{i=S} \\ &+ \sum_{j=\{d,c\}} (\Omega_{j,t-1} (v_{j,t} + P_{j,t} \Pi_{j,t}) - \Omega_{j,t} v_{j,t}) \cdot \mathbb{1}_{i=S} \end{aligned}$$

$$d_{i,t} = z_{i,t} + (1 - \delta) d_{i,t-1}$$

▷ non-HtM transversality condition:  $\lim_{k \rightarrow \infty} \prod_{h=0}^k (R_{t+h})^{-1} B_{t+k} = 0$

► Optimality conditions

► Preferences

# Firms

- ▷ Two industries:  $j = \{c, d\}$
- ▷ Split the production process, two stages
  - A perfectly competitive final good producer aggregates ...
  - ... intermediate varieties supplied by monopolistically competitive firms
- ▷ Labor-only technology
- ▷ NK optimal subsidy removes steady-states markup distortions (Debortoli and Galí, 2024)
- ▷ Sector-specific NK Phillips Curve (Calvo, 1983)

$$\hat{\pi}_{j,t} = \beta \mathbb{E}_t [\hat{\pi}_{j,t+1}] + (1 - \theta_j) (1 - \beta \theta_j) (\theta_j)^{-1} \widehat{mc}_{j,t}$$

- $mc_{j,t}$ : real marginal cost of intermediate firms in sector  $j$
- $(1 - \theta_j)$ : probability of price setting of intermediate firms in sector  $j$

# Counterpart of savings & monetary policy

- ▷ Exogenous and constant **non-durables endowment**  $\omega_c$ , no labor supply
- ▷ Maximizes utility from non-durable consumption  $c_{x,t}$ , using **bond holdings**  $B_{x,t}$ 
  - Balances the non-HtM's saving position
- ▷ To correct saving **imbalances**:  $R_{x,t+1} = R_{t+1} e^{-\phi_B B_{x,t}}$  (Schmitt-Grohé and Uribe, 2003)
- ▷ The **policy instrument** adjusts to sectoral inflation and a non-systematic disturbance

$$R_{t+1} = \bar{R}^{1-\rho_R} R_t^{\rho_R} e^{\iota_{m,t}} \prod_{j=\{d,c\}} (\pi_{j,t})^{\phi_j}$$

- $\phi_j > 1$ : sector-specific **reaction coefficient**
  - $\iota_{m,t}$ : AR(1) **monetary policy shock**
- ▷ Equal **after-profit transfers** across households

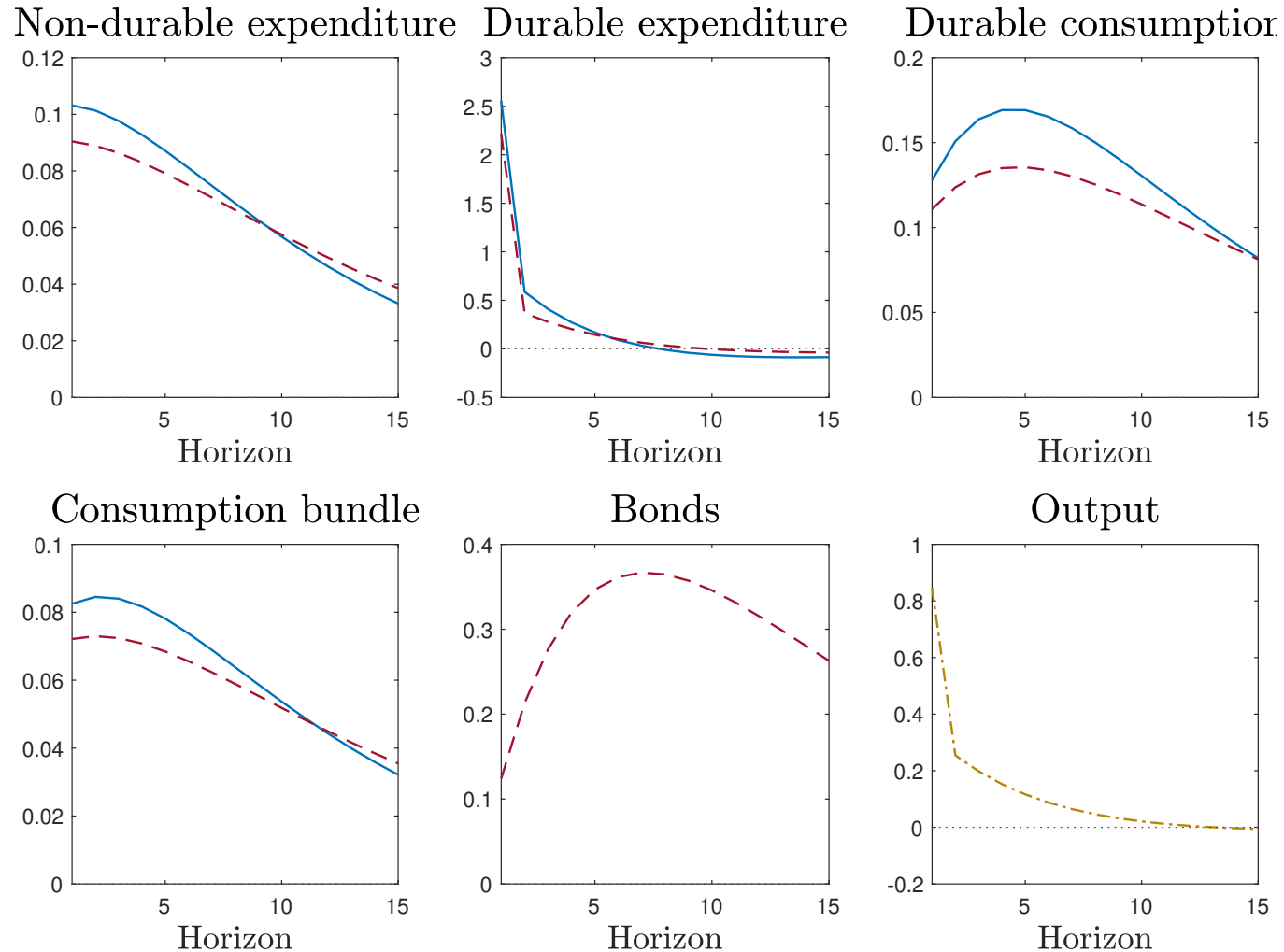
► **Market clearing conditions**

# Calibration

| Parameter  | Value | Source                 | Description                                             |
|------------|-------|------------------------|---------------------------------------------------------|
| $\beta$    | 0.99  | Beraja and Wolf (2021) | Households' discount factor                             |
| $\delta$   | 0.05  | Barsky et al. (2016)   | Rate of depreciation of durables                        |
| $\kappa$   | 0.00  | Beraja and Wolf (2021) | Governs the convexity of durables' adjustment costs     |
| $\sigma$   | 1.00  | Beraja and Wolf (2021) | Inverse of the intratemporal elasticity of substitution |
| $\zeta$    | 1.00  | Beraja and Wolf (2021) | Inverse of the intertemporal elasticity of substitution |
| $\eta$     | 0.50  | Bachmann et al. (2024) | Inverse of the Frisch labor supply elasticity           |
| $\chi$     | 10.0  | Bachmann et al. (2024) | Households' relative labor utility cost                 |
| $\phi$     | 0.70  | Barsky et al. (2016)   | Steady-state share of non-durable consumption           |
| $\theta$   | 0.75  | Bachmann et al. (2024) | Calvo's price stickiness in each sector.                |
| $\phi_j$   | 1.50  | Auclert et al. (2020)  | Reaction to sectoral inflation                          |
| $\rho_m$   | 0.83  | Auclert et al. (2020)  | Persistence of monetary policy shocks                   |
| $\sigma_m$ | 0.57  | Auclert et al. (2020)  | Standard deviation of monetary policy shocks            |



# Monetary policy shock | Result 1: HtM & non-HtM

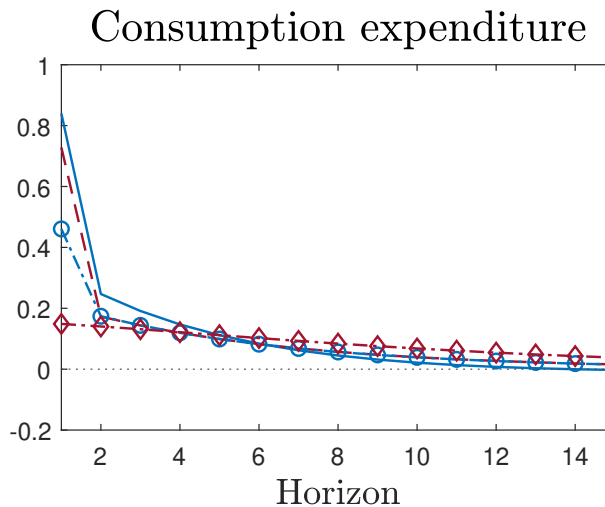
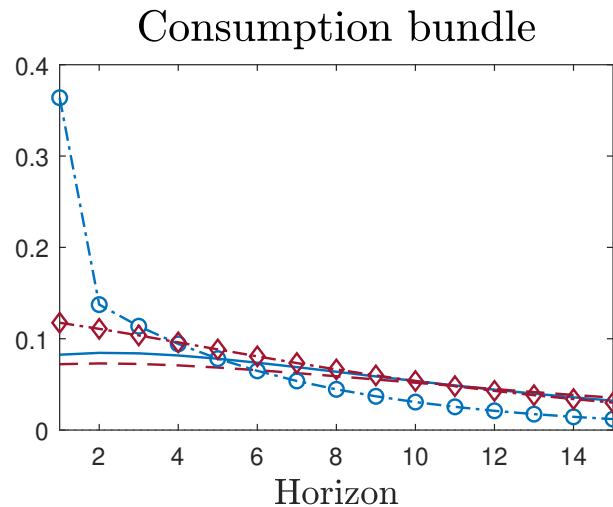
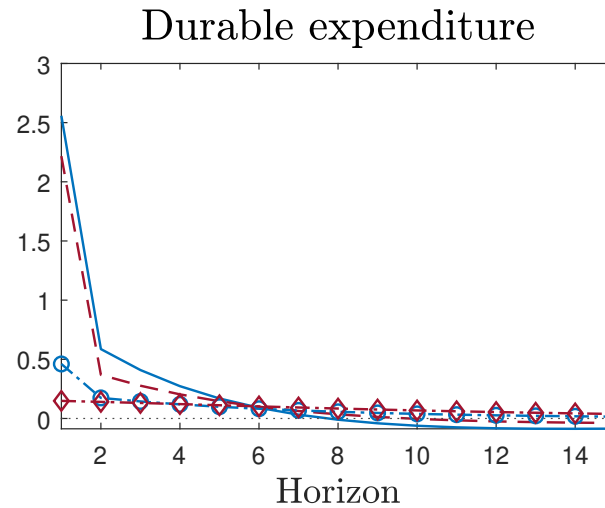
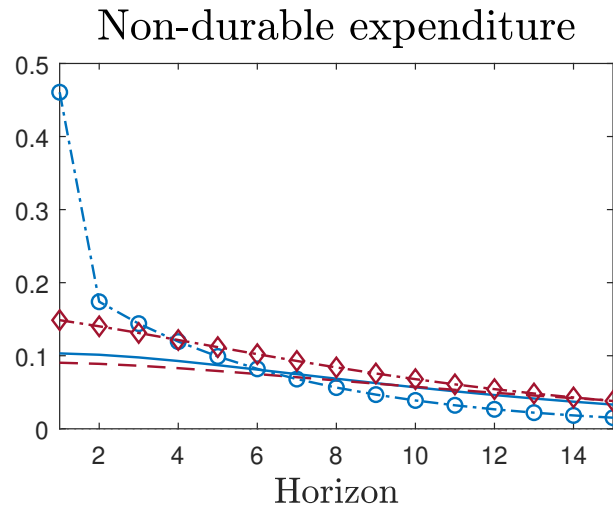


— HtM  
- - non-HtM

- Durables **substitute** for financial assets in smoothing consumption
- Durables loosen the **consumption-income** link

► All figures

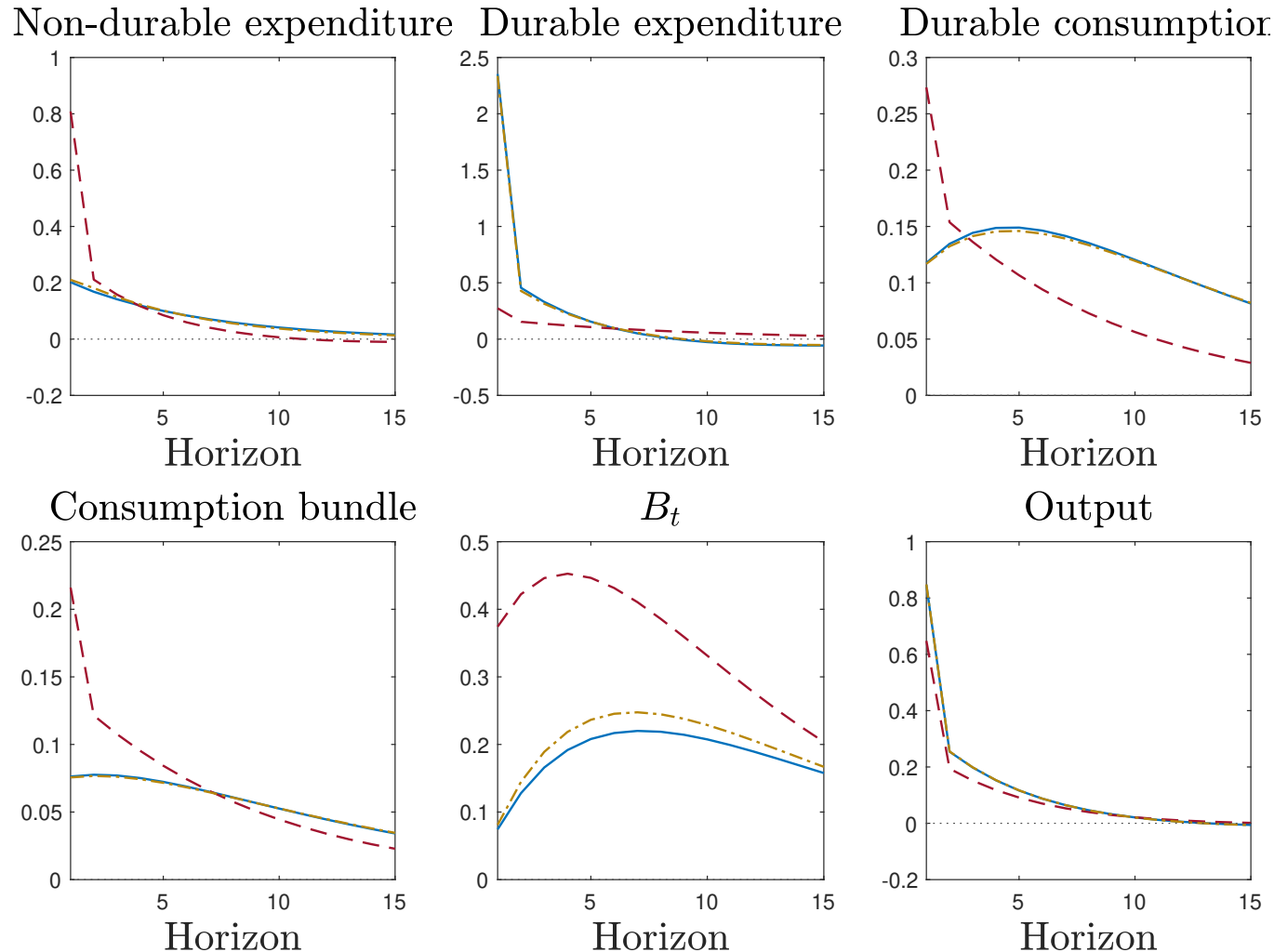
# Monetary policy shock | Result 2: TANK+D & TANK



- HtM, TANK+D
- - non-HtM, TANK+D
- -○- HtM, TANK
- -◇- non-HtM, TANK

- Sharp reaction on durable spending, **front-loading** to smooth consumption
- Spending becomes **more volatile**, yet consumption is **stabilized**

# Monetary policy shock | Result 3: TANK+D & RANK



— TANK+D  
- - TANK  
- . RANK

- The **forward-looking HtM** makes TANK+D to closely resemble RANK
- Durables **smooth** consumption, **dampening** households heterogeneity

► All figures

# Conclusions

- ▷ Durables introduce a forward-looking dimension for the HtM household
- ▷ The forward-looking behavior of the HtM household is relevant for individual & aggregate outcomes
- ▷ We provide empirical validation of the forward-looking HtM households
- ▷ Integrating durables into TANK models dampens the role of household heterogeneity for aggregate results

# References I

- Mark Aguiar, Mark Bilts, and Corina Boar. Who are the hand-to-mouth? *Review of Economic Studies*, 92(3):1293–1340, 2025.
- Emrehan Aktug and Orhan Erem Atesagaoglu. Inflation expectations and household spending in high inflation: Evidence from a randomized control trial. *Unpublished*, 2024.
- Fernando Alvarez, Robert E. Lucas, and Warren E. Weber. Interest rates and inflation. *American Economic Review*, 91(2):219–225, 2001.
- Philippe Andrade, Erwan Gautier, and Eric Mengus. What matters in households' inflation expectations? *Journal of Monetary Economics*, 138:50–68, 2023.
- Olivier Armantier, Scott Nelson, Giorgio Topa, Wilbert Van der Klaauw, and Basit Zafar. The price is right: Updating inflation expectations in a randomized price information experiment. *The Review of Economics and Statistics*, 56(2):503–523, 2016.
- Olivier Armantier, Giorgio Topa, Wilbert Van der Klaauw, and Basit Zafar. An overview of the survey of consumer expectations. *Economic Policy Review*, 23(2):51–72, 2017.
- Orazio P. Attanasio. Chapter 11 consumption. *Handbook of Macroeconomics*, 1(B):741–812, 1999.
- Adrien Auclert, Matthew Rognlie, and Ludwig Straub. Micro jumps, macro humps: Monetary policy and business cycles in an estimated HANK model. *NBER Working Paper Series*, 26647, 2020.

# References II

- Rüdiger Bachmann, Tim O. Berg, and Eric R. Sims. Inflation expectations and readiness to spend: Cross-sectional evidence. *American Economic Journal: Economic Policy*, 7(1):1–35, 2015.
- Rüediger Bachmann, Benjamin Born, Olga Goldfayn-Frank, Georgi Kocharkov, Ralph Luetticke, and Michael Weber. A temporary vat cut as unconventional fiscal policy. *NBER Working Paper Series*, 2024.
- Robert Barsky, Christopher L. House, Christoph Boehm, and Miles S. Kimball. Monetary policy and durable goods. *Unpublished*, 2016.
- Robert B. Barsky, Christopher L. House, and Miles S. Kimball. Sticky-price models and durable goods. *American Economic Review*, 97(3):984–998, 2007.
- Martín Beraja and Christian K. Wolf. Demand composition and the strength of recoveries. *NBER Working Paper Series*, Working Paper No. 29304, 2021.
- Florin O. Bilbiie. Limited asset market participation, monetary policy and (inverted) aggregate demand logic. *Journal of Economic Theory*, 140(1):162–196, 2008.
- Florin O. Bilbiie. The new keynesian cross. *Journal of Monetary Economics*, 114:90–108, 2020.
- Forin O. Bilbiie, Tommaso Monacelli, and Roberto Perotti. Public debt and redistribution with borrowing constraints. *The Economic Journal*, 123(566):F64–F98, 2012.
- Ricardo Caballero. Durable goods: An explanation for their slow adjustment. *Journal of Political Economy*, 101(2), 1993.

# References III

- Guillermo A. Calvo. Staggered prices in a utility-maximizing framework. *American Economic Review*, 12(3):383–398, 1983.
- John Y. Campbell and N. Gregory Mankiw. Consumption, income, and interest rates: Reinterpreting the time series evidence. In *NBER Macroeconomics Annual 1989, Volume 4*, NBER Chapters, pages 185–246. National Bureau of Economic Research, Inc, 1989.
- Bernardo Candia. Inflation expectations and household spending: Different patterns in low and high-inflation settings. *Unpublished*, 2024.
- Alberto Cavallo, Guillermo Cruces, and Ricardo Pérez-Truglia. Inflation expectations, learning, and supermarket prices: Evidence from survey experiments. *American Economic Journal: Macroeconomics*, 9(3):1–35, 2017.
- Michael P. Clements. Explanations of the inconsistencies in survey respondents’ forecasts. *European Economic Review*, 54(4):536–549, 2010.
- Michael P. Clements. Probability distributions or point predictions? Survey forecasts of US output growth and inflation. *International Journal of Forecasting*, 30(1):99–117, 2014.
- Olivier Coibion, Yuriy Gorodnichenko, and Michael Weber. Monetary policy communications and their effects on household inflation expectations. *Journal of Political Economy*, 130(6):1537–1584, 2022.

# References IV

- Olivier Coibion, Dimitris Georgarakos, Yuriy Gorodnichenko, and Maarten Van Rooij. How does consumption respond to news about inflation? Field evidence from a randomized control trial. *AEJ Macroeconomics*, 15(3):109–152, 2023.
- Olivier Coibion, Dimitris Georgarakos, Yuriy Gorodnichenko, Geoff Kenny, and Michael Weber. The effect of macroeconomic uncertainty on household spending. *American Economic Review*, 114(3):645–677, 2024.
- Francesco D’Acunto, Ulrike Malmendier, Juan Ospina, and Michael Weber. Exposure to grocery prices and inflation expectations. *Journal of Political Economy*, 129(5):1615–1639, 2021.
- Davide Debortoli and Jordi Galí. Heterogeneity and aggregate fluctuations: Insights from TANK models. *NBER Working Paper Series*, Working Paper No. 32557, 2024.
- Alexander M. Dietrich, Keith Kuester, Gernot J. Müller, and Raphael S. Schoenle. News and uncertainty about covid-19: Survey evidence and short-run economic impact. *Journal of Monetary Economics*, 129:S35–S51, 2022.
- Jordi Galí, J. David López-Salido, and Javier Vallés. Understanding the effects of government spending on consumption. *Journal of European Economic Association*, 5(1):227–270, 2007.
- Ingar Haaland, Christopher Roth, and Johannes Wohlfart. Designing information provision experiments. *Journal of Economic Literature*, 61(1):3–40, 2023.
- Ina Hajdini, Edward S. Knotek, John Leer, Mathieu Pedemonte, Robert Rich, and Raphael S. Schoenle. Indirect consumer inflation expectations. *Journal of Monetary Economics*, 145(103568), 2024.



# References V

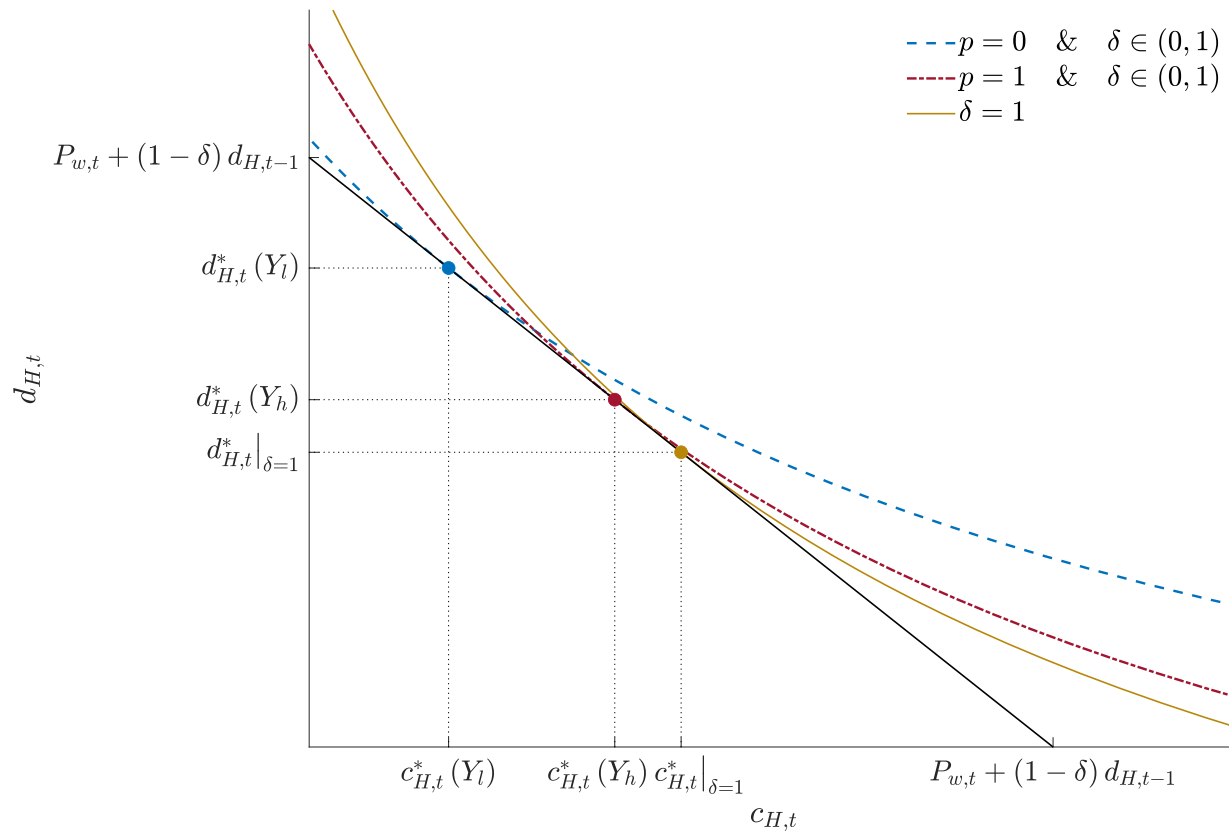
- Janet Hua Jiang, Rupal Kamdar, Kelin Lu, and Daniela Puzzello. How do households respond to expected inflation? an investigation on transmission mechanisms. *CAEPR Working Papers*, 004, 2024.
- Greg Kaplan and Giovanni L. Violante. A model of the consumption response to fiscal stimulus payments. *Econometrica*, 82(4):1199–1239, 2014.
- Greg Kaplan, Giovanni L. Violante, and Justin Weidner. The wealthy hand-to-mouth. *Brooking Papers on Economic Activity*, Spring, 2014.
- Robert G. King, Charles I. Plosser, and Sergio T. Rebelo. Production, growth and business cycles. *Journal of Monetary Economics*, 21:195–232, 1988.
- Edward S. Knotek II, Raphael S. Schoenle, Alexander M. Dietrich, Keith Kuester, Gernot J. Müller, Kristian Ove R. Myrseth, and Michael Weber. Consumers and covid-19: A real-time survey. *Federal Reserve Bank of Cleveland, Economic Commentary*(2020-08), 2020.
- Christos Kotsogiannis and Plutarchos Sakellaris. Mpcs estimated from tax lottery, survey and administrative data. *Unpublished*, 2025.
- Alisdair McKay and Johannes F. Wieland. Lumpy durable consumption demand and the limited ammunition of monetary policy. *Econometrica*, 89(6):2717–2749, 2021.
- Alisdair McKay and Johannes F. Wieland. Forward guidance and durable goods demand. *American Economic Review: Insights*, 4(1):106–122, 2022.

# References VI

- Alisdair McKay, Emi Nakamura, and Jón Steinsson. The power of forward guidance revisited. *American Economic Review*, 106(10):3133–3158, 2016.
- Christopher Roth and Johannes Wohlfart. How do expectations about the macroeconomy affect personal expectations and behavior? *The Review of Economics and Statistics*, 102(4):731–748, 2020.
- Christopher Roth, Mirko Wiederholt, and Johannes Wholfart. The effects of monetary policy: Theory with measured expectations. *Unpublished*, 2023.
- Stephanie Schmitt-Grohé and Martín Uribe. Closing small open economy models. *Journal of International Economics*, 61:163–185, 2003.
- Philip Schnorpfel, Michael Weber, and Andreas Hackethal. Households’ response to the wealth effects of inflation. *NBER Working Paper Series*, Working Paper No. 31672, 2023.
- Stephen P. Zeldes. Consumption and liquidity constraints: An empirical investigation. *Journal of Political Economy*, 97(2):305–346, 1989.

# Optimal HtM consumption

HtM consumption in period  $t$  using the stripped-down model



- $\delta = 1$  : allocations are independent of future states of the world,  $p$  is irrelevant
- $\delta \in (0, 1)$  flattens the IC, more when anticipating higher marginal utility

# Optimal consumption responses

## Lemma 1

Given a **one-time, isolated and exogenous** change in the expected inflation  $\mathbb{E}_t [\pi_{j,t+1}]$  of the nominal price  $P_{j,t} / j = \{d, c, w\}$ , **optimal consumption responses** satisfy

$$\begin{aligned}\frac{\partial z_{i,t}}{\partial \mathbb{E}_t [\pi_{j,t+1}]} &= \mathcal{D}_i \left( \frac{\partial \mathbb{E}_t [q_{t+1}]}{\partial \mathbb{E}_t [\pi_{j,t+1}]}, \frac{\partial \mathbb{E}_t [w_{t+1}]}{\partial \mathbb{E}_t [\pi_{j,t+1}]}, \frac{\partial \mathbb{E}_t [r_{t+1}]}{\partial \mathbb{E}_t [\pi_{j,t+1}]} \cdot \mathbb{1}_{i=s} \mid \Theta \right) \\ \frac{\partial c_{i,t}}{\partial \mathbb{E}_t [\pi_{j,t+1}]} &= \mathcal{C}_i \left( \frac{\partial \mathbb{E}_t [q_{t+1}]}{\partial \mathbb{E}_t [\pi_{j,t+1}]}, \frac{\partial \mathbb{E}_t [w_{t+1}]}{\partial \mathbb{E}_t [\pi_{j,t+1}]}, \frac{\partial \mathbb{E}_t [r_{t+1}]}{\partial \mathbb{E}_t [\pi_{j,t+1}]} \cdot \mathbb{1}_{i=s} \mid \Theta \right)\end{aligned}$$

where  $i = \{H, S\}$ , and  $\mathcal{D}_i(\bullet), \mathcal{C}_i(\bullet) : \mathbb{R}^3 \rightarrow \mathbb{R}$ .

- Policy function: **household-specific**
- Responses: hinge on changes in **expected relative prices**
- HtM: **no sensitivity** to changes in expected **real interest rate**

# HtM candidates | Survey questions

## Question 1 : *paycheck-to-paycheck*

QH.1 Thinking about your current household's situation, how strongly do you agree or disagree with the statement: "My household lives *paycheck-to-paycheck*"?

- ☐ I strongly agree.
- ☐ I somewhat agree.
- ☐ I neither agree nor disagree.
- ☐ I somewhat disagree.
- ☐ I strongly disagree.

## Question 2 : *borrow*

QH.2 If your household needed to *borrow one week's worth of your monthly household income*, how easy or difficult would it be for your household to come up with the money?

- ☐ It would be extremely difficult.
- ☐ It would be relatively difficult.
- ☐ It would be neither easy nor difficult.
- ☐ It would be relatively easy.
- ☐ It would be extremely easy.

# HtM candidates | Survey questions

## Question 3 : *savings*

QH.3 Which fraction of your income do you [save](#)?

- ☐ I spend all of my income each month.
- ☐ I spend more money than I earn, and I often use credit cards or other loans to supplement my monthly income.
- ☐ I save around 10% of my monthly income.
- ☐ I save around 25% of my monthly income.
- ☐ I save at least 50% of my monthly income.

## Question 4 : *weeks*

QH.4 Suppose your household loses its primary source of income today. [How long can your household cover its weekly expenses](#) with your household's total money in checking and savings accounts or other liquid financial assets such as cash, stocks, government and corporate bonds, and mutual and money market funds? Please give your best guess.

- If my household loses its primary source of income today, we can cover our expenses for \_\_\_\_\_ week(s) using the money in our checking and saving accounts plus other liquid financial assets.

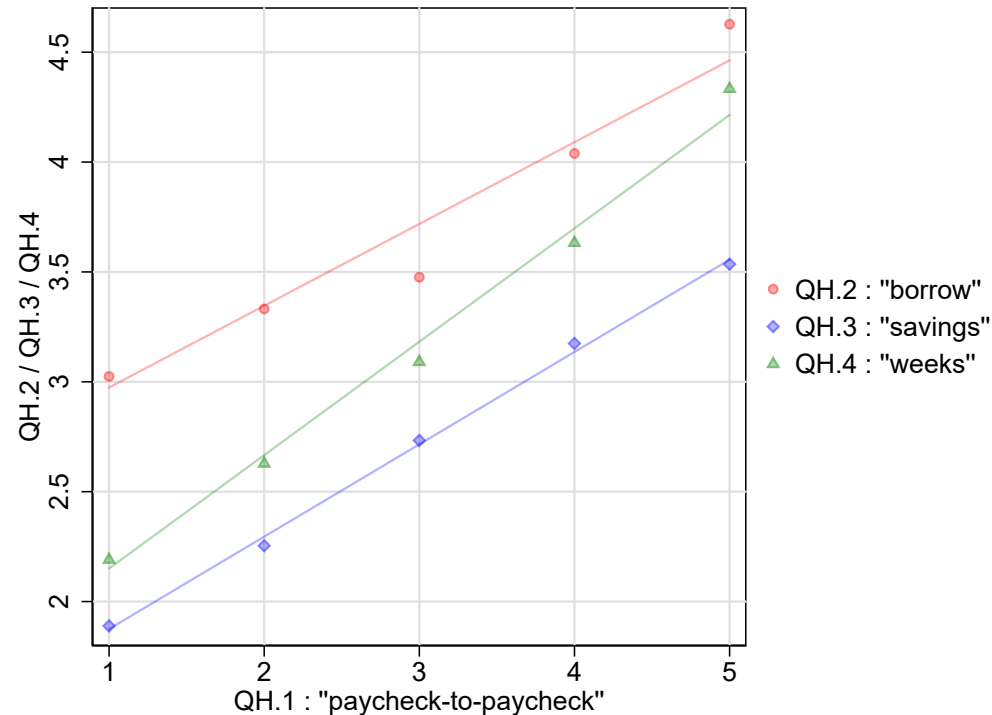
# Mean values by HtM status

Mean value of demographic and socioeconomic variables by HtM status

|                    | Full sample                    | non-HtM                        | HtM                            |
|--------------------|--------------------------------|--------------------------------|--------------------------------|
| Age                | 49.1                           | 49.8                           | 48.1                           |
| Female             | 0.50                           | 0.44                           | 0.58                           |
| Number of children | 1.50                           | 1.55                           | 1.44                           |
| Education          | Some college,<br>but no degree | Some college,<br>but no degree | Some college,<br>but no degree |
| Income             | \$35,000-\$49,999              | \$50,000-\$99,999              | \$35,000-\$49,999              |
| Completion time    | 12.68                          | 12.60                          | 12.80                          |
| Observations       | 5,529                          | 3,231                          | 2,298                          |

# Strong correlation & internal consistency

Relationship between the answers to the different HtM questions



**Note:** The figure presents binned scatter plots between the different HtM questions and the question labeled as "*paycheck-to-paycheck*" (QH.1), with the latter plotted on the x-axis and the remaining questions on the y-axis. The number of bins used in each plot is 5.

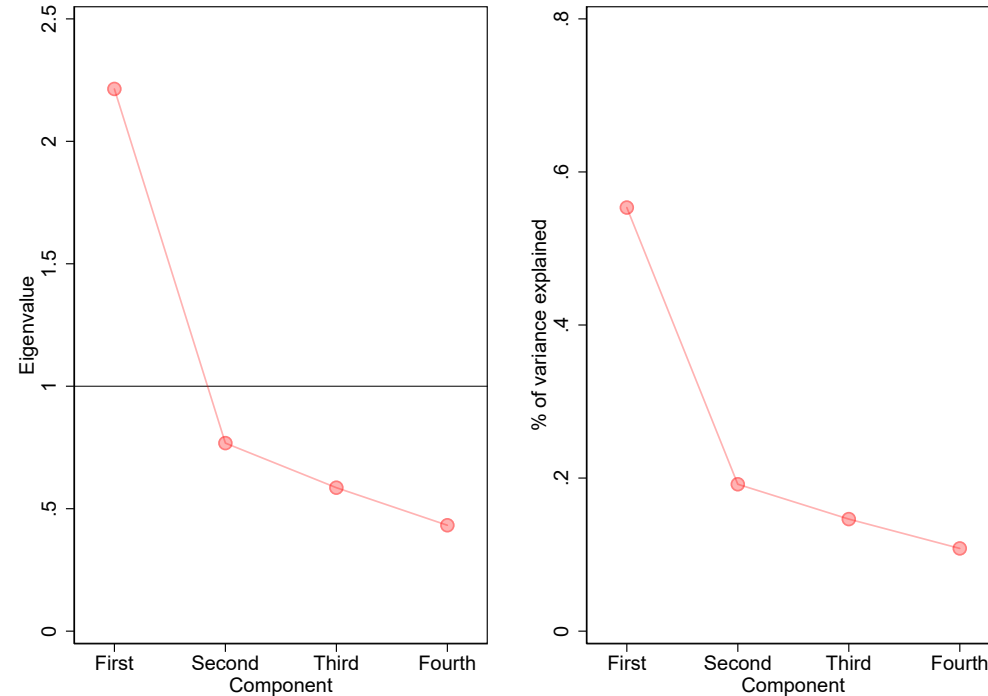
- Lower values represent a higher probability of being an HtM respondent

[Return](#)



# Dominant & significant share of variance

Eigenvalue and Variance explained of each principal component



**Note:** The figure provides two scree plots summarizing the results of the PCA. The left panel plots the eigenvalue associated with each principal component, while the right panel illustrates the proportion of total variance explained by each principal component.

# Sample balanced across treatment arms

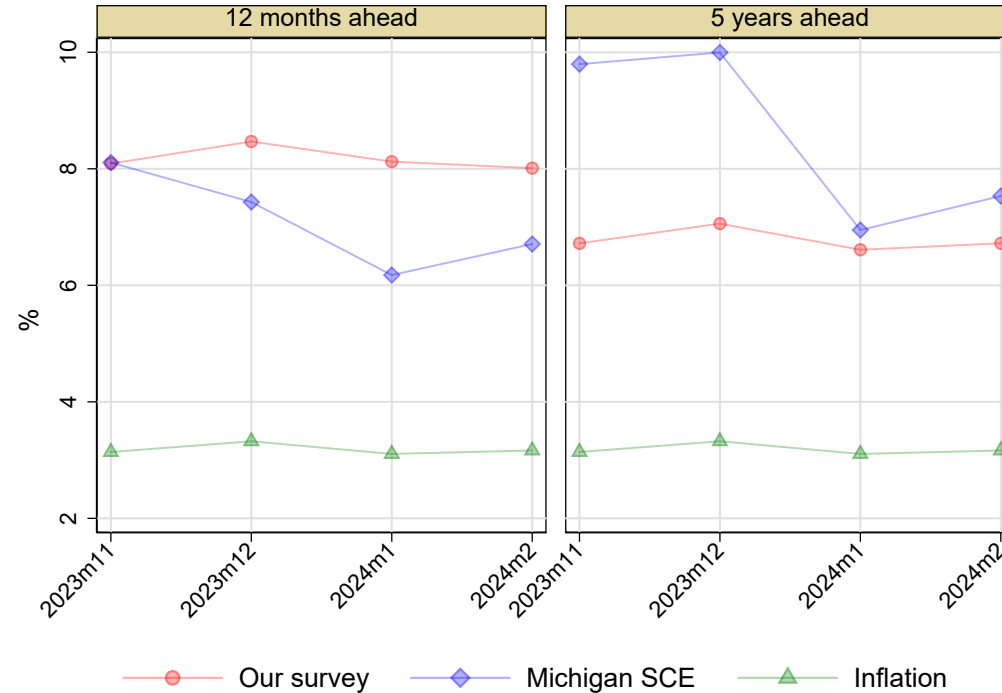
Balance test on pre-treatment covariates

|                       | T-test or $\chi^2$ -test p-value |        |       |         |        |        |
|-----------------------|----------------------------------|--------|-------|---------|--------|--------|
|                       | C & T1                           | C & T2 | C & P | T1 & T2 | T1 & P | T2 & P |
| Age                   | 0.92                             | 0.82   | 0.23  | 0.92    | 0.34   | 0.40   |
| Gender                | 0.54                             | 0.37   | 0.99  | 0.81    | 0.59   | 0.44   |
| Education level       | 0.55                             | 0.14   | 0.82  | 0.13    | 0.39   | 0.75   |
| Civil status          | 0.22                             | 0.20   | 0.24  | 0.70    | 0.34   | 0.64   |
| Household net income  | 0.09                             | 0.32   | 0.07  | 0.05    | 0.10   | 0.05   |
| Number of children    | 0.58                             | 0.91   | 0.67  | 0.70    | 0.94   | 0.77   |
| Political affiliation | 0.51                             | 0.95   | 0.74  | 0.27    | 0.77   | 0.45   |
| English fluency       | 0.46                             | 0.08   | 0.39  | 0.56    | 0.21   | 0.05   |

**Note:** The columns in the table report p-values from balance tests conducted across treatment groups for different demographic and socioeconomic variables measured before treatment implementation. We conducted a t-test or a  $\chi^2$ -test depending on the variable’s nature. C denotes the passive control group, P corresponds to the placebo group, and T1 and T2 are the first and second treatment groups. The second row in the table provides information about the comparison groups in the analysis performed in each column.

# Prior & posterior beliefs

## Inflation expectations and actual inflation rate



**Note:** The figure presents the time series of the averages of our collected data and the raw data from the University of Michigan's Survey of Consumers. The series plotted are the annual inflation expectations 12 months and 5 years ahead, alongside the actual inflation rate. Our collected inflation expectations data have been trimmed at the weekly frequency by excluding the top and bottom 2.5% of the distribution.

# Descriptive statistics by survey & HtM status

## Summary statistics on respondents' inflation expectations

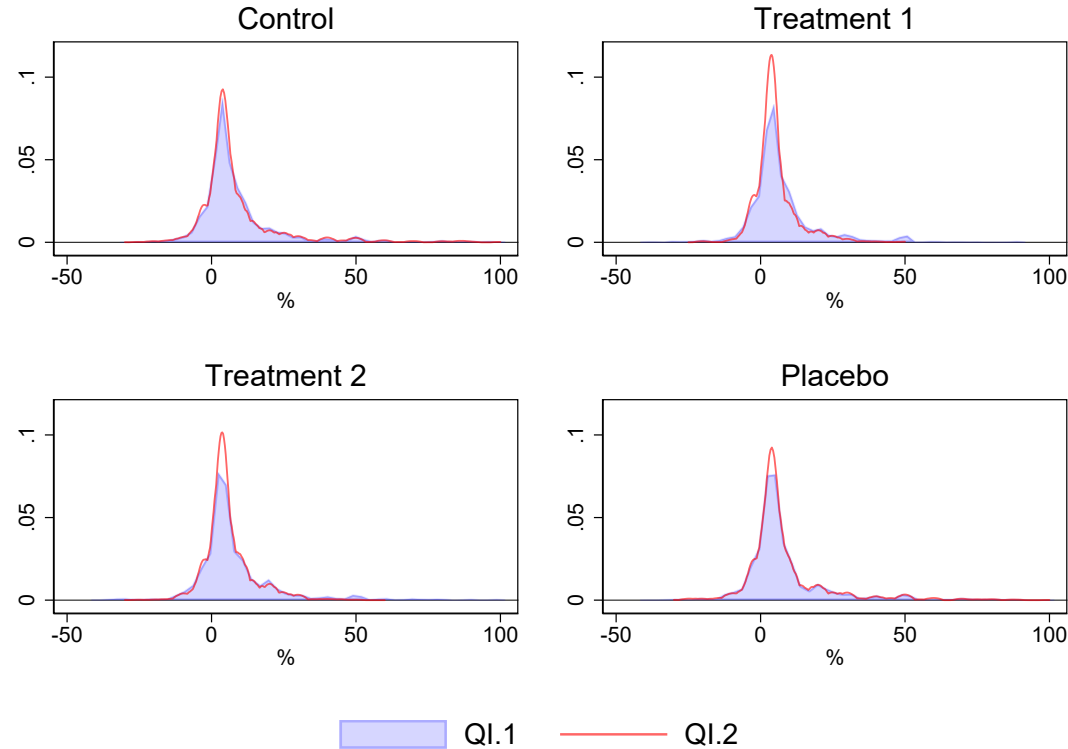
| Inflation expectations | Michigan SCE |        | Full sample |        | HtM  |        | non-HtM |        |
|------------------------|--------------|--------|-------------|--------|------|--------|---------|--------|
|                        | Mean         | Median | Mean        | Median | Mean | Median | Mean    | Median |
| 12 months ahead        | 6.8          | 6.7    | 8.2         | 5.0    | 8.7  | 5.0    | 7.8     | 4.0    |
| 5 years ahead          | 8.3          | 7.5    | 6.8         | 4.0    | 6.9  | 5.0    | 6.6     | 4.0    |

**Note:** The table reports descriptive statistics of the answers to the prior (Q1.1) and posterior (Q1.2) questions using our collected data and the raw data from the University of Michigan's Survey of Consumers. Our collected inflation expectations data have been trimmed at the weekly frequency by excluding the top and bottom 2.5% of the distribution.

[Return](#)

# Treatments | Validation

Kernel density of inflation expectations by treatment status



- Posterior densities of T1 & T2: bunching on treatment value & larger kurtosis
- Suggestive evidence of successful treatments affecting beliefs

# Treatments | Validation

## Treatment effects on medium-run inflation expectations

|                                                | (1)                   | (2)                   |                                        |
|------------------------------------------------|-----------------------|-----------------------|----------------------------------------|
| Variable <sub>j</sub>                          |                       | Variable <sub>j</sub> | Variable <sub>j</sub> × H <sub>j</sub> |
| <i>prior<sub>j</sub></i>                       | 0.856***<br>(0.012)   | 0.857***<br>(0.008)   | −0.033<br>(0.023)                      |
| <i>treat<sub>j,1</sub></i>                     | 1.798***<br>(0.197)   | 1.540***<br>(0.276)   | 0.284<br>(0.289)                       |
| <i>treat<sub>j,1</sub> × prior<sub>j</sub></i> | − 0.486***<br>(0.018) | − 0.424***<br>(0.035) | −0.041<br>(0.037)                      |
| <i>treat<sub>i,2</sub></i>                     | 1.894***<br>(0.204)   | 1.526***<br>(0.226)   | 0.926***<br>(0.324)                    |
| <i>treat<sub>j,2</sub> × prior<sub>j</sub></i> | − 0.459***<br>(0.017) | − 0.442***<br>(0.031) | −0.010<br>(0.035)                      |
| <i>treat<sub>i,3</sub></i>                     | 0.149<br>(0.194)      | −0.036<br>(0.221)     | 0.424<br>(0.278)                       |
| <i>treat<sub>j,3</sub> × prior<sub>j</sub></i> | 0.001<br>(0.016)      | −0.000<br>(0.019)     | 0.005<br>(0.035)                       |
| Observations                                   | 5,529                 | 5,529                 |                                        |
| R <sup>2</sup>                                 | 0.792                 | 0.793                 |                                        |

**Note:** The dependent variable is the posterior belief measure *post<sub>j</sub>*. The omitted category is the passive control group. Robust standard errors are reported in parentheses. The set of fixed effects includes week and U.S. state fixed effects. \*, \*\*, and \*\*\* denote significance at the 10%, 5%, and 1% level, respectively. The inflation expectations data have been trimmed at the weekly frequency by excluding the top and bottom 2.5% of the distribution.

# Heterogeneous effects by HtM status | 2SLS

▷ First-stage:

$$\begin{aligned}
 \text{post}_j \times (1 - H_j) &= \alpha^S + \mathbf{X}_j (\boldsymbol{\beta}^S)' + \eta^S \times \text{prior}_j + \sum_{\tau=\{1,2\}} \left[ \gamma_{\tau,1,S}^S (\text{treat}_{j,\tau} \times (1 - H_j)) + \gamma_{\tau,1,H}^S (\text{treat}_{j,\tau} \times H_j) \right. \\
 &\quad \left. + \gamma_{\tau,2,S}^S (\text{treat}_{j,\tau} \times \text{prior}_j \times (1 - H_j)) + \gamma_{\tau,2,H}^S (\text{treat}_{j,\tau} \times \text{prior}_j \times H_j) \right] + v_j^S \\
 \text{post}_j \times H_j &= \alpha^H + \mathbf{X}_j (\boldsymbol{\beta}^H)' + \eta^H \times \text{prior}_j + \sum_{\tau=\{1,2\}} \left[ \gamma_{\tau,1,S}^H (\text{treat}_{j,\tau} \times (1 - H_j)) + \gamma_{\tau,1,H}^H (\text{treat}_{j,\tau} \times H_j) \right. \\
 &\quad \left. + \gamma_{\tau,2,S}^H (\text{treat}_{j,\tau} \times \text{prior}_j \times (1 - H_j)) + \gamma_{\tau,2,H}^H (\text{treat}_{j,\tau} \times \text{prior}_j \times H_j) \right] + v_j^H
 \end{aligned}$$

▷ Second-stage:

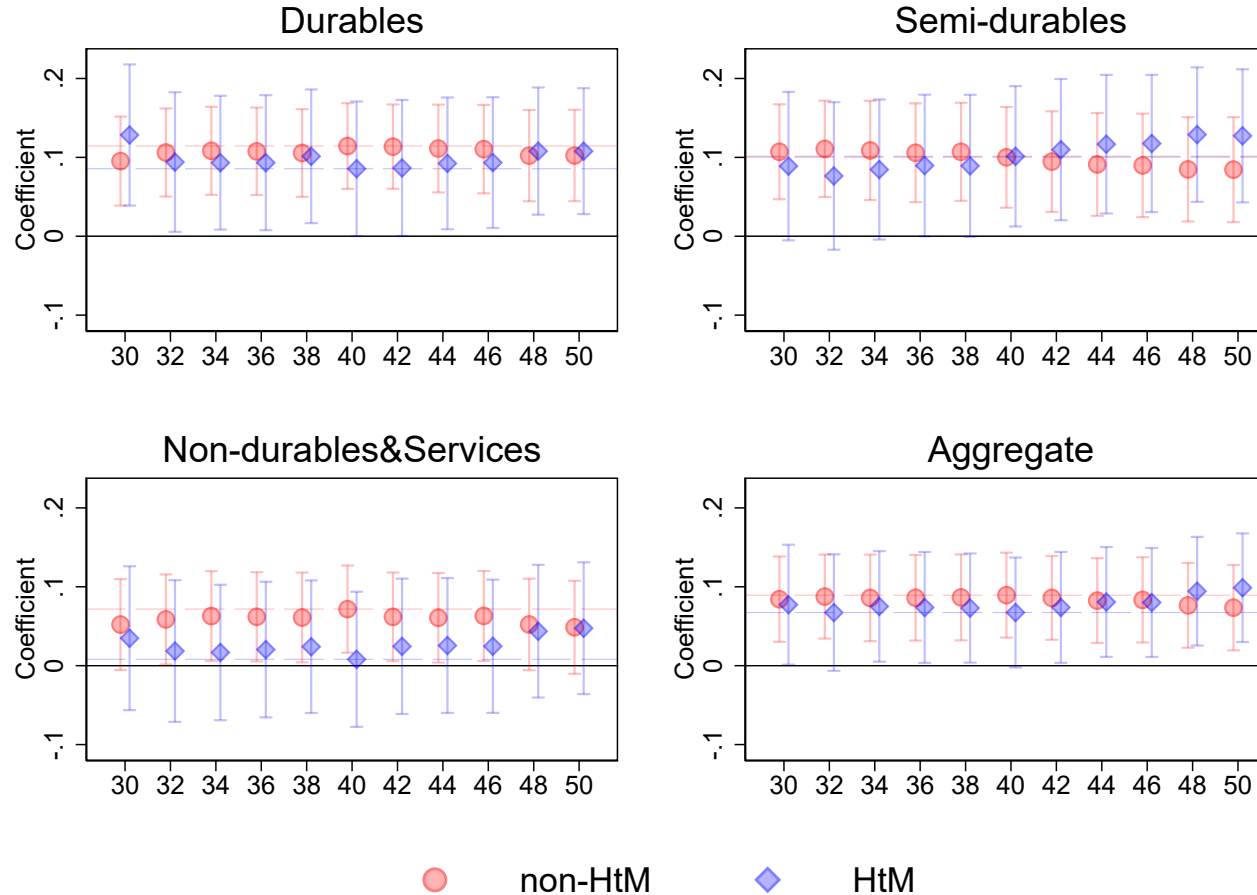
$$\% \Delta c_{j,k,h}^e = \alpha_{k,h} + \mathbf{X}_j \boldsymbol{\beta}_{k,h}' + \eta_{k,h} \times \text{prior}_j + \theta_{k,h}^S \times \widehat{\text{post}_j \times (1 - H_j)} + \theta_{k,h}^H \times \widehat{\text{post}_j \times H_j} + \varepsilon_{j,k,h}$$

- **Key coefficients:**  $\theta_{k,h}^i$  captures the expectations' causal effect for  $i = \{H, S\}$

[Return](#)

# Stability of the inflation expectations coefficients

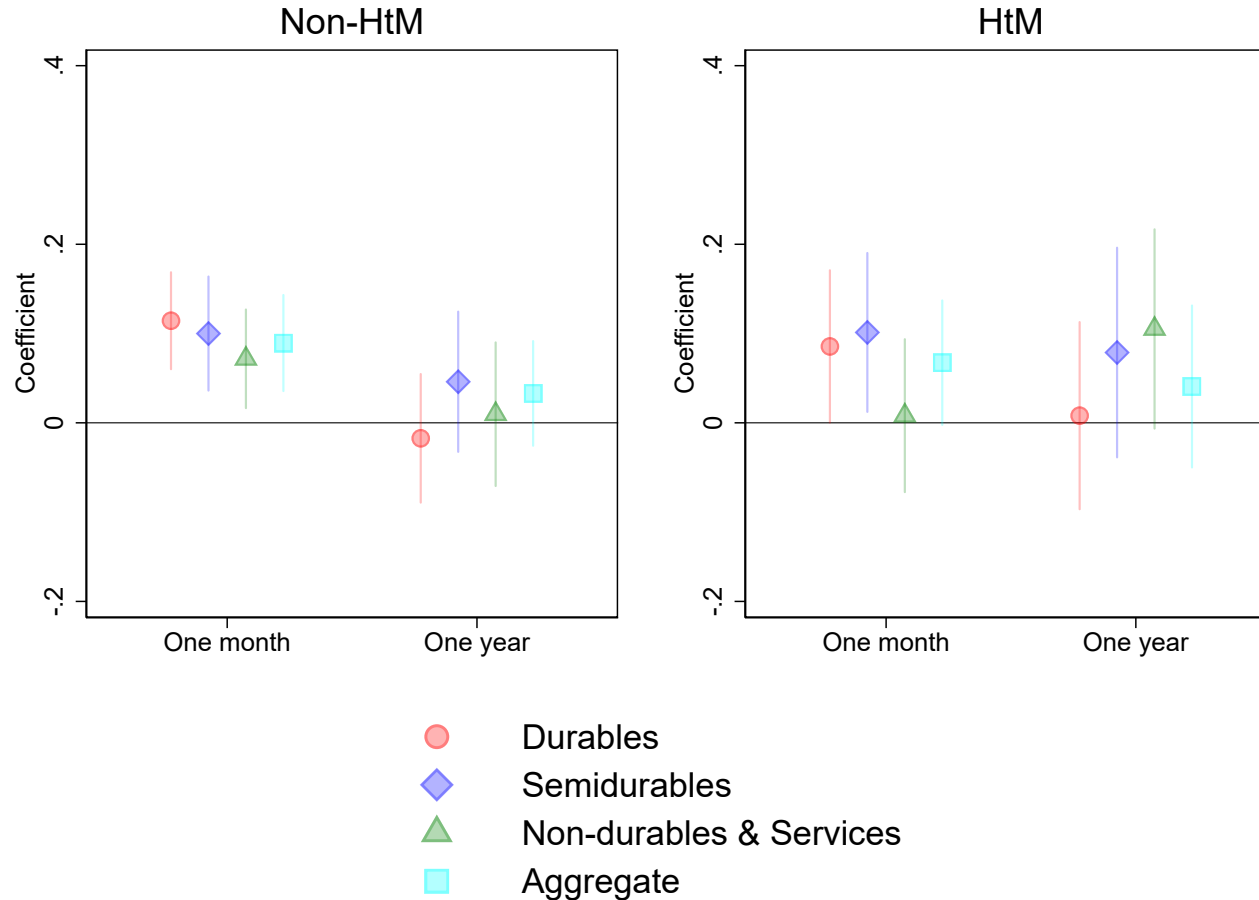
Estimated coefficients with alternative HtM cutoff





# Evidence | Higher aggregate inflation expectations, pooled

Aggregate inflation expectations changes on consumption by HtM status



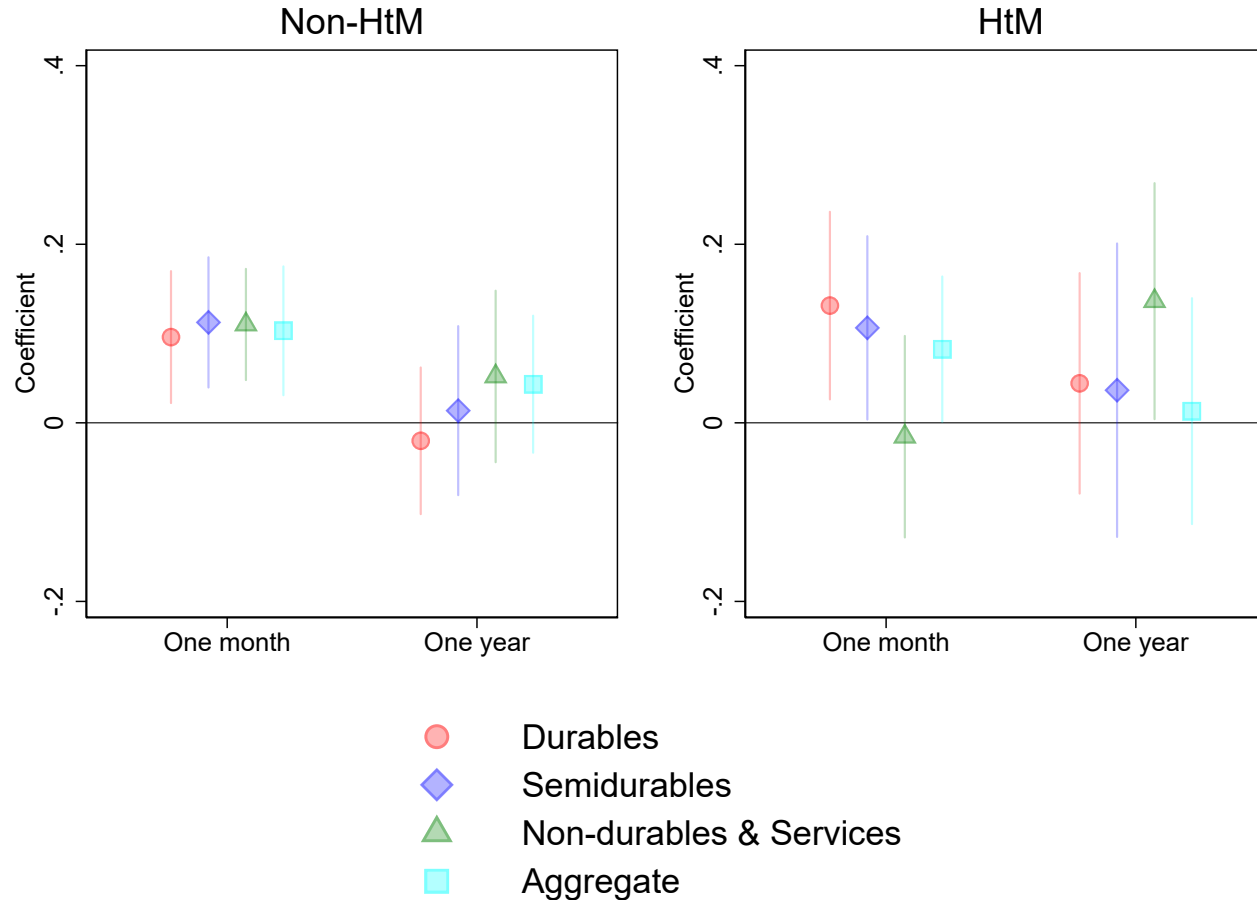
# Evidence | Higher aggregate inflation expectations, pooled

## Effects of durable inflation expectations on consumption

|                           | % $\Delta c_{j,k,h}^e$ : Intended consumption growth on category $k$ at horizon $h$ |                   |                     |                  |                           |                  |                     |                  |
|---------------------------|-------------------------------------------------------------------------------------|-------------------|---------------------|------------------|---------------------------|------------------|---------------------|------------------|
|                           | Durables                                                                            |                   | Semi-durables       |                  | Non-durables and Services |                  | Aggregate           |                  |
|                           | (1)                                                                                 | (2)               | (3)                 | (4)              | (5)                       | (6)              | (7)                 | (8)              |
| $post_j \times (1 - H_j)$ | 0.114***<br>(0.033)                                                                 | -0.017<br>(0.043) | 0.100***<br>(0.038) | 0.046<br>(0.047) | 0.071**<br>(0.033)        | 0.009<br>(0.048) | 0.089***<br>(0.032) | 0.033<br>(0.035) |
| $post_j \times H_j$       | 0.085*<br>(0.051)                                                                   | 0.008<br>(0.063)  | 0.101*<br>(0.054)   | 0.078<br>(0.071) | 0.008<br>(0.052)          | 0.105<br>(0.067) | 0.067<br>(0.042)    | 0.040<br>(0.055) |
| Horizon                   | One month                                                                           | One year          | One month           | One year         | One month                 | One year         | One month           | One year         |
| Observations              | 5,099                                                                               | 5,087             | 5,095               | 5,106            | 4,946                     | 4,953            | 5,147               | 5,160            |
| Fixed-effects             | Yes                                                                                 | Yes               | Yes                 | Yes              | Yes                       | Yes              | Yes                 | Yes              |
| Controls                  | Yes                                                                                 | Yes               | Yes                 | Yes              | Yes                       | Yes              | Yes                 | Yes              |
| F-stat, non-HtM           | 179.9                                                                               | 176.9             | 174.0               | 178.5            | 171.6                     | 173.7            | 183.5               | 184.2            |
| F-stat, HtM               | 154.8                                                                               | 157.4             | 158.5               | 159.0            | 145.1                     | 151.3            | 154.0               | 155.9            |
| R <sup>2</sup>            | 0.134                                                                               | 0.069             | 0.139               | 0.079            | 0.150                     | 0.077            | 0.223               | 0.205            |

# Evidence | Higher durable inflation expectations, T2

Durable inflation expectations changes on consumption by HtM status



# Evidence | Higher durable inflation expectations, T2

Durable inflation expectations changes on consumption by HtM status

|                           | % $\Delta c_{j,k,h}^e$ : Intended consumption growth on category $k$ at horizon $h$ |                   |                    |                  |                           |                   |                    |                  |
|---------------------------|-------------------------------------------------------------------------------------|-------------------|--------------------|------------------|---------------------------|-------------------|--------------------|------------------|
|                           | Durables                                                                            |                   | Semi-durables      |                  | Non-durables and Services |                   | Aggregate          |                  |
|                           | (1)                                                                                 | (2)               | (3)                | (4)              | (5)                       | (6)               | (7)                | (8)              |
| $post_j \times (1 - H_j)$ | 0.096**<br>(0.044)                                                                  | -0.020<br>(0.049) | 0.112**<br>(0.044) | 0.013<br>(0.057) | 0.110**<br>(0.037)        | 0.052<br>(0.058)  | 0.103**<br>(0.043) | 0.043<br>(0.046) |
| $post_j \times H_j$       | 0.131**<br>(0.063)                                                                  | 0.044<br>(0.075)  | 0.106*<br>(0.062)  | 0.036<br>(0.099) | -0.015<br>(0.037)         | 0.136*<br>(0.058) | 0.082*<br>(0.049)  | 0.013<br>(0.076) |
| Horizon                   | One month                                                                           | One year          | One month          | One year         | One month                 | One year          | One month          | One year         |
| Observations              | 3,788                                                                               | 3,776             | 3,796              | 3,810            | 3,689                     | 3,690             | 3,837              | 3,843            |
| Fixed-effects             | Yes                                                                                 | Yes               | Yes                | Yes              | Yes                       | Yes               | Yes                | Yes              |
| Controls                  | Yes                                                                                 | Yes               | Yes                | Yes              | Yes                       | Yes               | Yes                | Yes              |
| F-stat, non-HtM           | 190.0                                                                               | 186.5             | 188.3              | 201.1            | 198.2                     | 197.1             | 208.3              | 203.7            |
| F-stat, HtM               | 164.4                                                                               | 164.4             | 167.7              | 169.7            | 155.9                     | 163.5             | 163.1              | 166.5            |
| R <sup>2</sup>            | 0.146                                                                               | 0.074             | 0.153              | 0.093            | 0.161                     | 0.094             | 0.224              | 0.204            |

# Households | Optimality conditions

▷ Two for HtM & non-HtM,  $i = \{H, S\}$

$$w_t = -\frac{mu_{i,t}^l}{mu_{i,t}^c}$$

$$mu_{i,t}^d = q_t mu_{i,t}^c \left(1 + \frac{\partial \psi_{i,t}}{\partial d_{i,t}}\right) - \beta \mathbb{E}_t \left[ q_{t+1} mu_{i,t+1}^c \left( (1 - \delta) \left(1 + \frac{\partial \psi_{i,t+1}}{\partial d_{i,t+1}}\right) - \frac{\partial \psi_{i,t+1}}{\partial d_{i,t}} \right) \right]$$

▷ Two extra conditions for **non-HtM**,  $i = S$

$$1 = \beta R_{t+1} \mathbb{E}_t \left[ mu_{S,t+1}^c (mu_{S,t}^c)^{-1} (\pi_{c,t+1})^{-1} \right]$$

$$v_{j,t} = \beta \mathbb{E}_t \left[ mu_{S,t+1}^c (mu_{S,t}^c)^{-1} (v_{j,t+1} + P_{j,t+1} \Pi_{j,t+1}) (\pi_{c,t+1})^{-1} \right]$$

# Households | Preferences

- ▷ Quadratic adjustment costs:  $\kappa \geq 0$  governs convexity (Beraja and Wolf, 2021)

$$\psi(d_{i,t}; d_{i,t-1}) = \frac{\kappa}{2} \left( \frac{d_{i,t}}{d_{i,t-1}} - 1 \right)^2 d_{i,t}$$

- ▷ King-Plosser-Rebelo type of preferences (King et al., 1988)

$$x(c_{i,t}, d_{i,t}) = \left[ \varrho^\sigma (c_{i,t})^{1-\sigma} + (1 - \varrho)^\sigma (d_{i,t})^{1-\sigma} \right]^{\frac{1}{1-\sigma}}$$
$$u(x_{i,t}, l_{i,t}) = \frac{(x_{i,t})^{1-\zeta}}{(1-\zeta)} - \chi \frac{(l_{i,t})^{1+\eta}}{(1+\eta)}$$

- $\sigma \in (0, 1)$ : inverse of the intratemporal elasticity of substitution
- $\zeta > 0$ : inverse of the intertemporal elasticity of substitution
- $\eta > 0$ : inverse of the Frisch labor supply elasticity
- $\varrho \in (0, 1)$ : preference parameter for non-durable goods

# Market clearing conditions

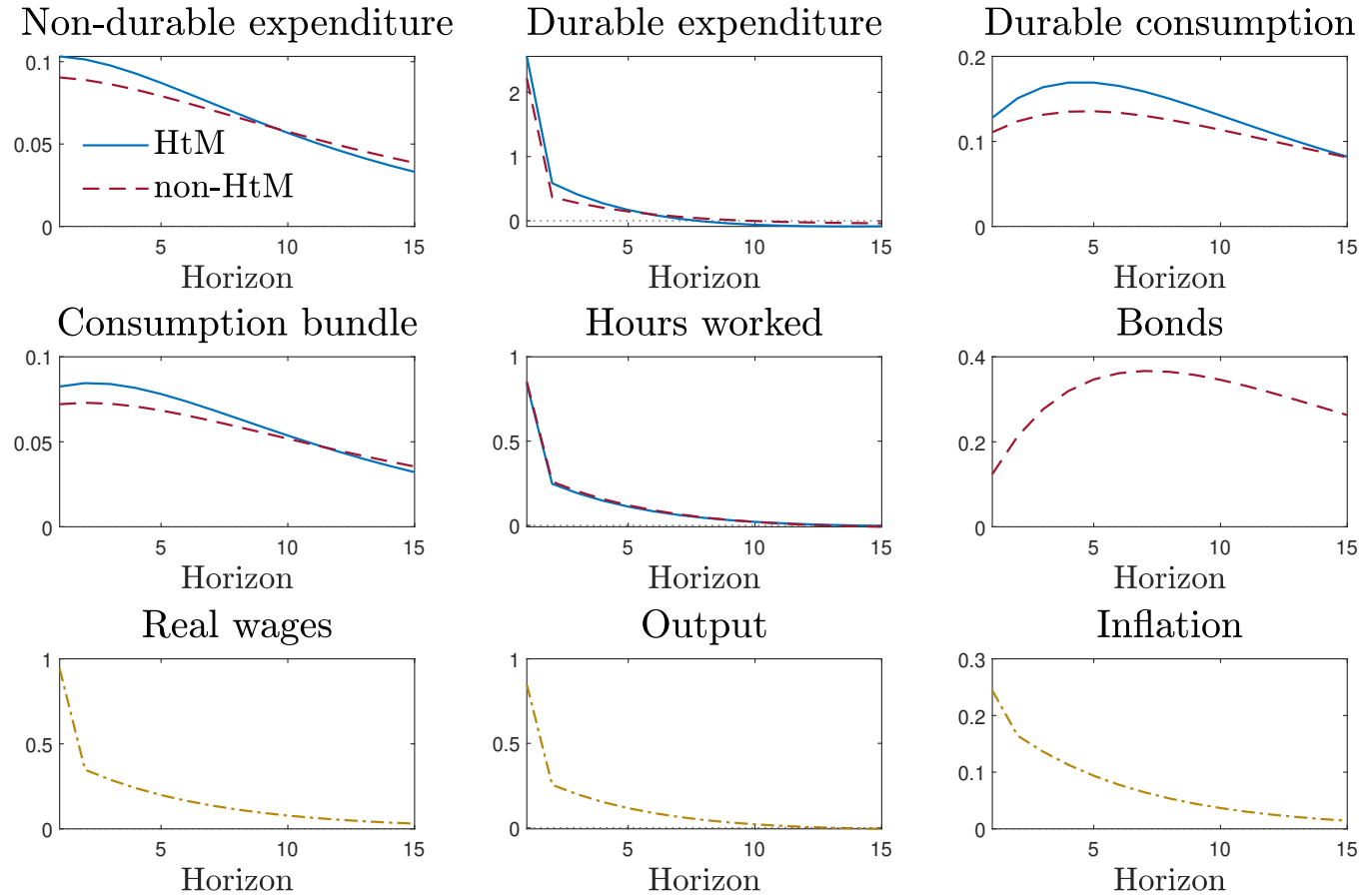
- ▷ Bonds market clears:  $B_{x,t} = \Lambda^s B_t$
- ▷ Equity market clears in each sector  $j = \{d, c\}$ :  $(\Lambda^s)^{-1} = \Omega_{j,t}$
- ▷ Labor market clears:

$$\sum_{i=\{H,S\}} \Lambda^i l_{i,j,t} = \int_{s_j \in \mathcal{F}_j} n_{s_j,t} ds_j = n_{j,t}$$
$$\sum_{i=\{H,S\}} \Lambda^i L_{i,t} = \sum_{j=\{d,c\}} \int_{s_j \in \mathcal{F}_j} n_{s_j,t} ds_j = \sum_{j=\{d,c\}} n_{j,t}$$

- ▷ Both goods markets clear:

$$y_{c,t} + \omega_c = \sum_{i=\{H,S\}} \Lambda^i c_{i,t} + c_{x,t}$$
$$y_{d,t} = \sum_{i=\{H,S\}} \Lambda^i [z_{i,t} + \psi(d_{i,t}; d_{i,t-1})]$$

# Expansionary monetary policy shock | HtM v/s non-HtM





# Expansionary monetary policy shock | TANK+D v/s RANK

